

Machine Learning

Lecture 1 — What Does It Mean to Learn?

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Fall semester · 4 credits · assessed by a pass or fail credit

Four things you will do

and two examinations on top of them

Lectures

ten weeks of ideas

Practicals

ten notebooks, not graded

Homework

six assignments, 30 points

Mini-project

one study, 25 points

How the grade is put together

100 points in total



100 points in total · at least 30 are needed before you may sit the final assessment

Every experiment you run has the same five steps



- **Experiment**
what you actually ran
- **Decision**
the choice you made, and why
- **Evidence**
the plot, table or number that supports it
- **Interpretation**
what that evidence means
- **Counterfactual**
what would change if one thing were different

the same five steps in every practical session, every assignment and the project

Where everything is, and two rules

Where to find things

The course website holds the syllabus, every module, the demonstrations and the policies.

dmytro-kuzmenko.github.io/ml-bsc-naukma

Notebooks open either on your own machine or in Google Colab.

Everything runs on an ordinary laptop processor.

Two rules to remember today

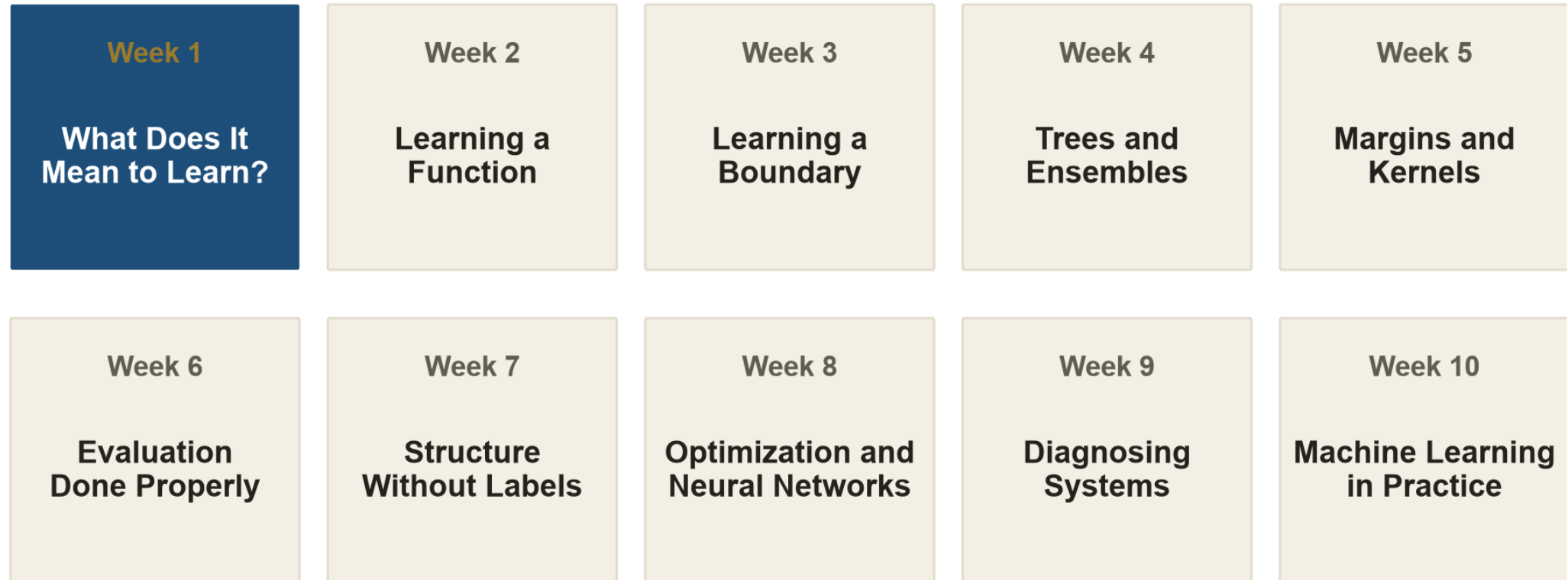
Artificial-intelligence tools are permitted for homework and for the project, provided you disclose how you used them.

They are prohibited during the midterm and the final examination.

There is no detection software. Understanding is verified by asking you about your own work.

Any submission may be selected for a short oral check.

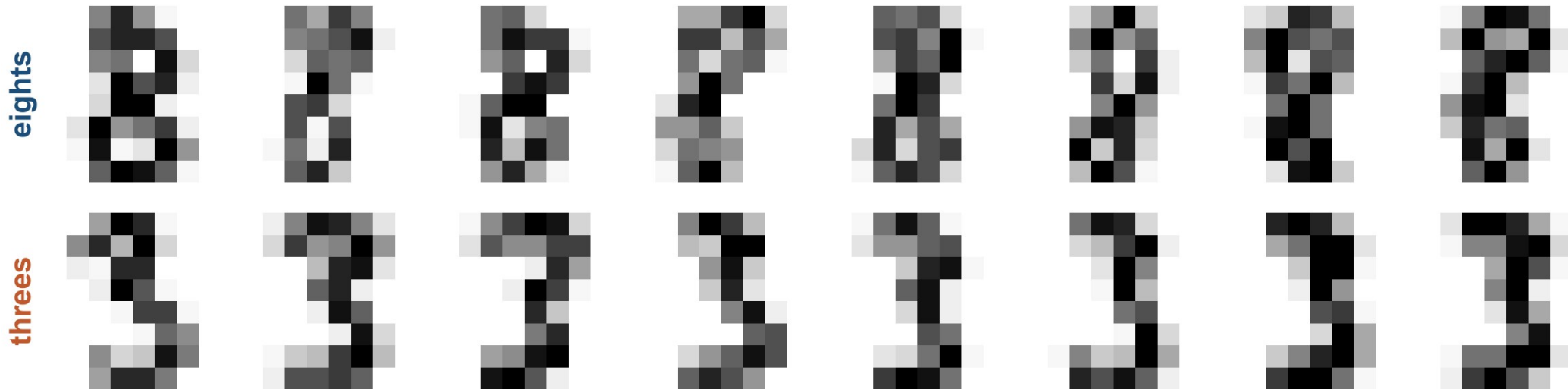
Ten weeks, and where today sits



How would you write a program that recognizes a handwritten digit?

Not train one. Write one, by hand, the way you would write any other program.

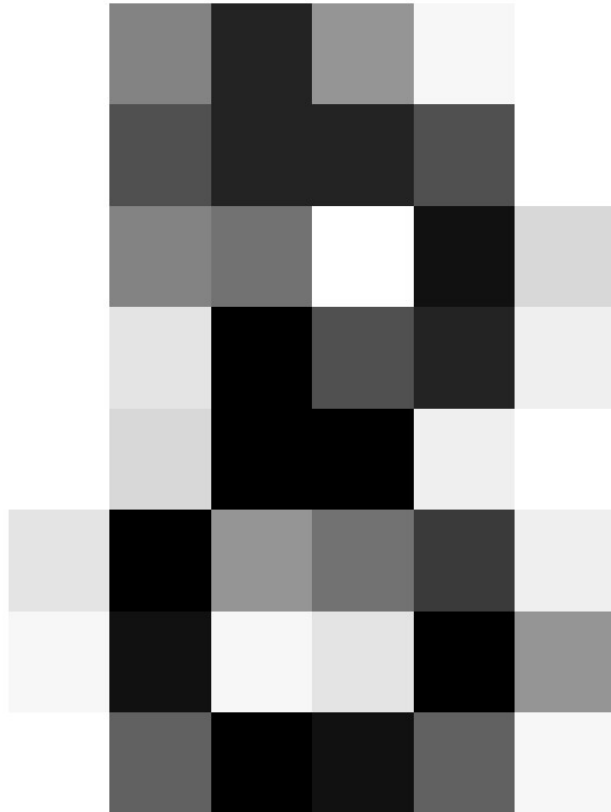
Every one of these is a handwritten digit



eight pixels by eight, from the handwritten-digits dataset in scikit-learn

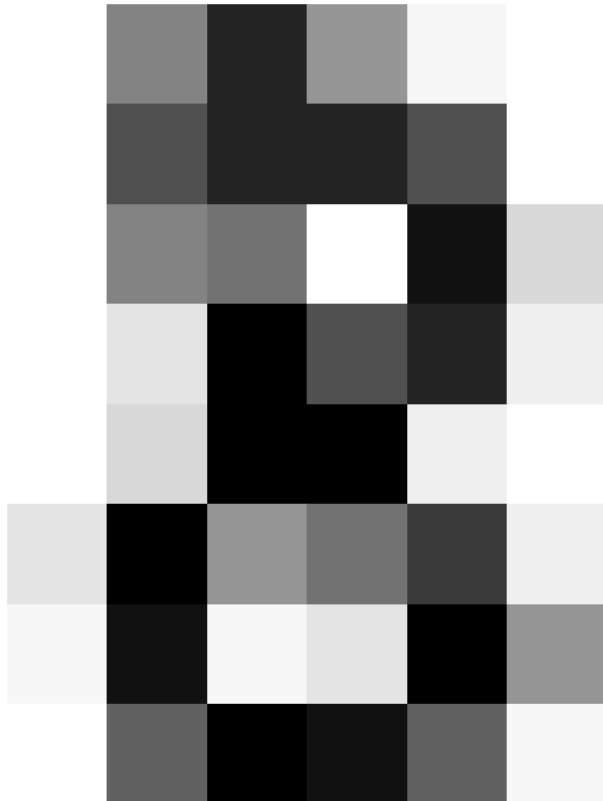
Try to write down the rule

this is an eight



Try to write down the rule

this is an eight



A rule for recognizing an eight

1. two closed loops, one above the other
2. the loops meet at a single crossing point
3. the whole shape is taller than it is wide

**Now write the rule for a badly drawn eight,
and for the next one, and for every one after that.**

Two ways to get a program

The usual way

a person writes the rules; the computer applies them



Two ways to get a program

The usual way

a person writes the rules; the computer applies them



The other way

a person supplies the answers; the computer writes the rules





A field of study that gives computers the ability to learn without being explicitly programmed.

Arthur Samuel, 1959, on the checkers program he had built at IBM

This is already running underneath ordinary life

Filtering spam

is this message unwanted?

Reading post codes

which digit is this?

Medical screening

which scan needs a specialist?

Machine translation

what does this sentence mean?

Detecting fraud

is this transaction unusual?

Recommendation

what would they watch next?

What this course is actually for

One sentence, and everything else in the semester follows from it

**formulate, train, evaluate, diagnose and explain
an ordinary machine-learning system.**

Not library syntax. Not one function call. Method: knowing what an experiment measures, and why its result can be trusted.

The rest of this lecture

- 1** What does it mean to learn?
- 2** What does the machine actually receive?
- 3** What does success mean?
- 4** Why is performance on the training data not enough?
- 5** Why the design of the evaluation decides everything
- 6** The shape of an experiment



What does it mean to learn?

Part 1 · the question the whole course rests on

What does it mean to learn?

Think of something you learned recently. How did you know you had learned it?

How would you teach a computer to tell spam from ordinary mail?

The rules people write

Rules a person might write

contains the word "free"

written entirely in capitals

The rules people write

Rules a person might write

contains the word "free"

written entirely in capitals

What goes wrong

- ➔ blocks a legitimate offer of a free trial
- ➔ blocks an excited message from a friend

The rules people write

Rules a person might write

contains the word "free"

written entirely in capitals

sender is on the block list

more than three exclamation marks

What goes wrong

- blocks a legitimate offer of a free trial
- blocks an excited message from a friend
- the sender changes address every week
- spammers simply use two

Every rule you add creates a new mistake, and the sender adapts faster than you can write them.

Rules are hard to write; examples are easy to collect

From writing the answer to writing the search for it

- Nobody can write down what spam is. Everybody can point at a message and say whether it is spam.
- What we do have is thousands of messages that somebody has already labeled.
- So we stop writing the rule and write a procedure that finds the rule in the examples.
- The machine is not given the answer. It is given the answers, and asked for the rule.

Learning, stated precisely

Tom Mitchell, Machine Learning, 1997

**A program learns from experience E , with respect to a task T and a measure of performance P ,
if its performance at T , as measured by P , improves with E .**

Three things you must be able to name

Every one of the three has to be stated before the claim means anything

***E* — the experience: the data the program is shown**

***T* — the task: what it is supposed to do**

***P* — the performance measure: how we decide it got better**

If you cannot name all three, no claim about learning can be checked.

The same three things, twice over

One that predicts a category, one that predicts a number

Filtering spam

E — a hundred thousand messages, each already marked as spam or not · T — decide which of the two a new message is · P — the share of new messages it gets right

Pricing an apartment

E — the record of what apartments sold for last year · T — estimate what this apartment is worth · P — how far the estimate falls from the price it eventually sells for

Notice that the task and the measure change together. The measure is chosen for the task, never the other way round.

Learning is performance that improves with experience

- We do not supply the answer. We supply the experience, and a way of measuring whether the answer got better.
- The same recipe covers spam, prices, medical images and recommendations, because none of them depends on what the data are about.
- Everything that follows is about making that measurement honest.

What does the machine actually receive?

Part 2 · samples, features and targets

What do we know about each example?

For a message: the text, the sender, the time it arrived. For an apartment: what else?

A dataset is a table

	size, m ²	rooms	floor	price
1	54	2	3	72 000
2	38	1	9	51 000
3	77	3	2	104 000
4	61	2	6	83 000
...	...	...	...	...

A dataset is a table

	size, m ²	rooms	floor	price
1	54	2	3	72 000
2	38	1	9	51 000
3	77	3	2	104 000
4	61	2	6	83 000
...	...	...	...	...

one row is one apartment: a single sample

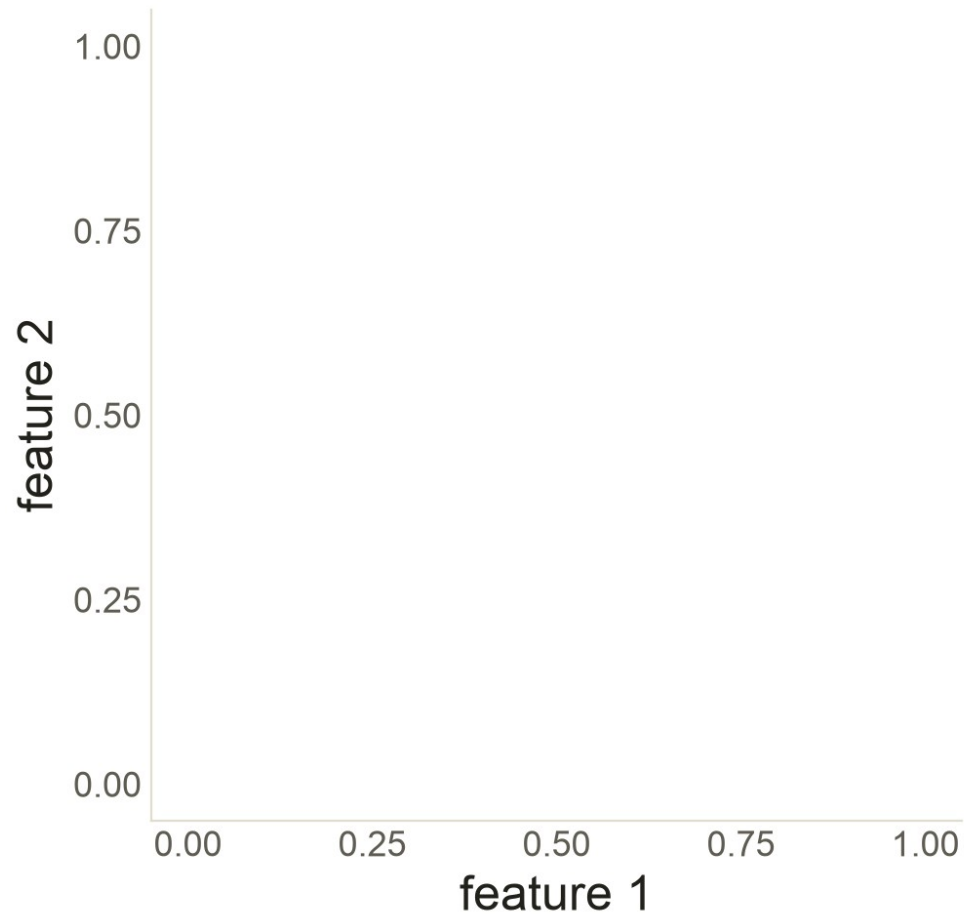
A dataset is a table

the answer we want to predict: the target
▼

	size, m ²	rooms	floor	price
1	54	2	3	72 000
2	38	1	9	51 000
3	77	3	2	104 000
4	61	2	6	83 000

...
one row is one apartment: a single sample

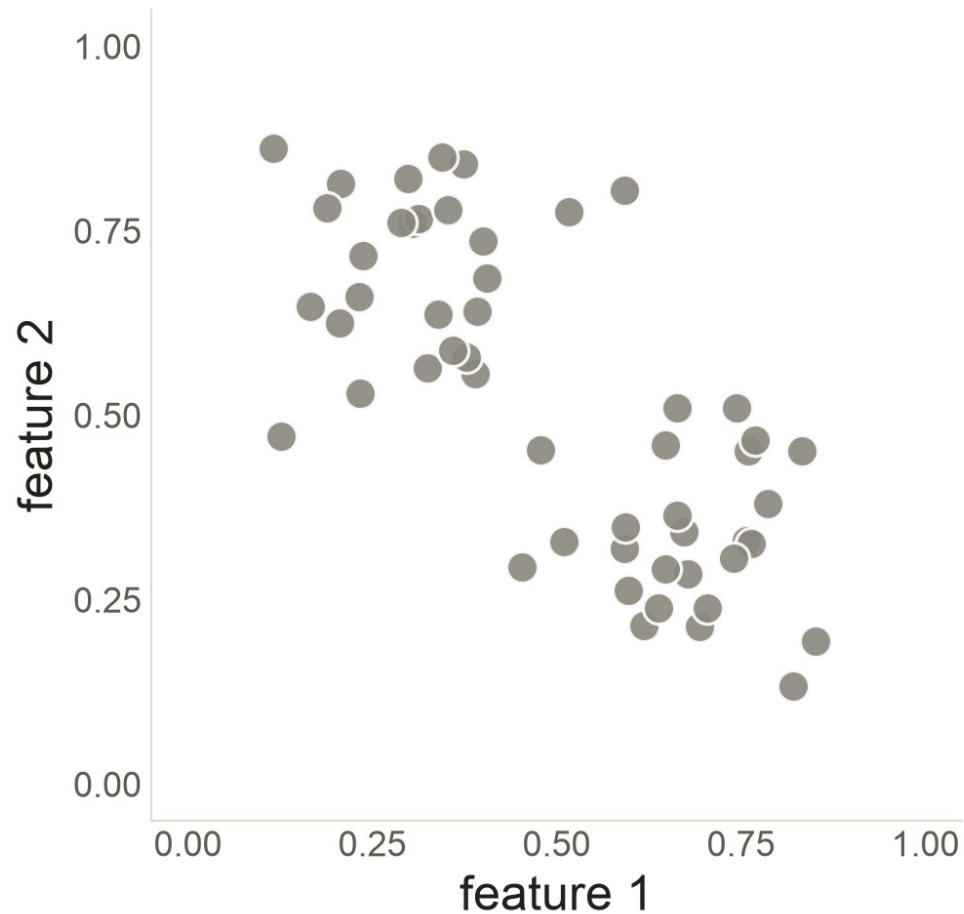
One sample is one point



two features, so two axes

nothing has been plotted yet

One sample is one point

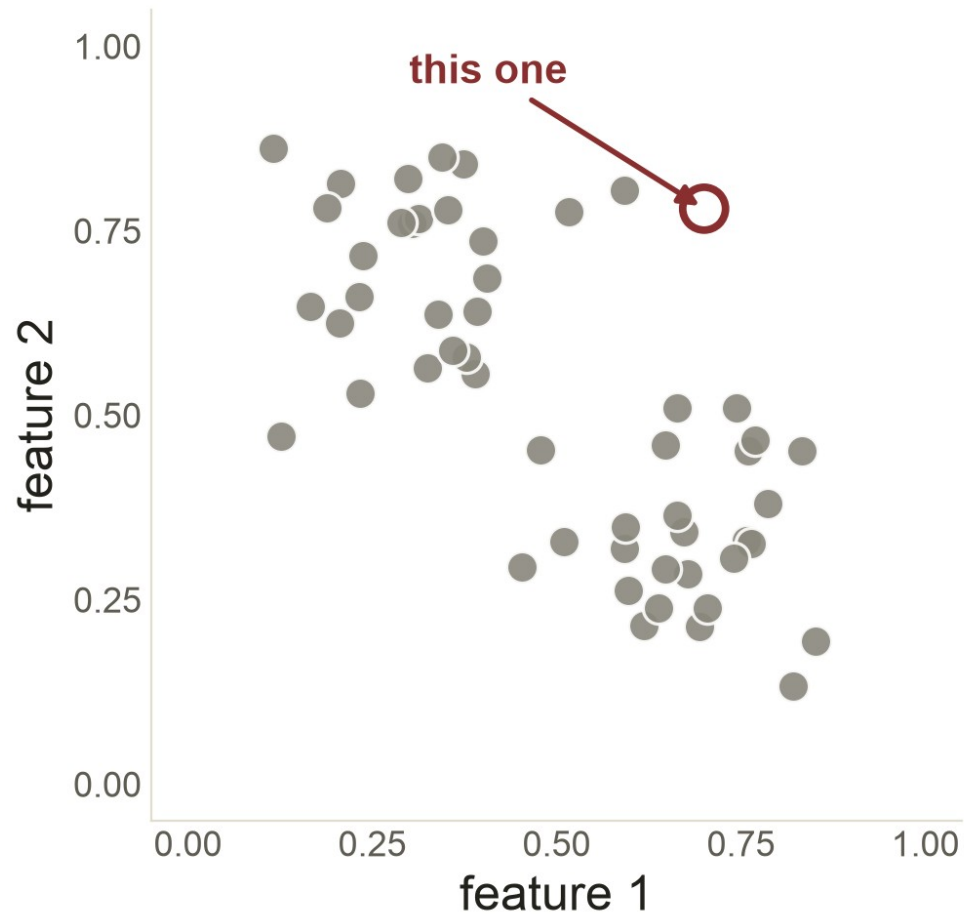


every sample is one dot

its position is what we measured

nothing else about it survives

One sample is one point



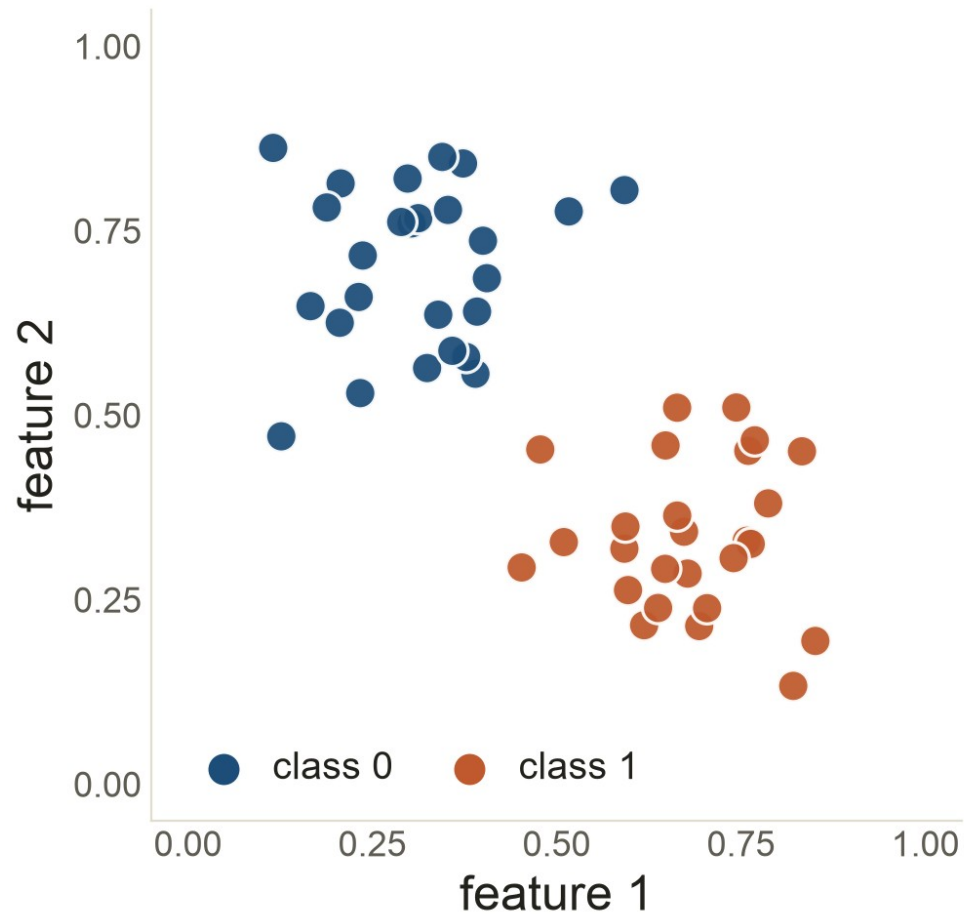
one dot is one apartment,

one message, one patient

dots that sit close together

are samples that resemble each other

One sample is one point



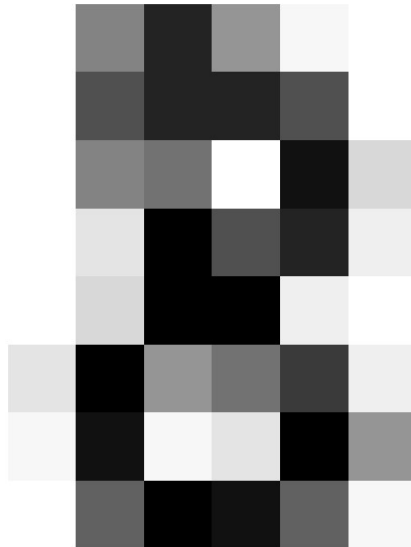
and now the answers

the two classes occupy
different regions of the space

**a classifier is a rule for deciding
which region a new dot falls in**

An image is a feature vector too

what a person sees



what the computer receives

0	0	9	14	8	1	0	0
0	0	12	14	14	12	0	0
0	0	9	10	0	15	4	0
0	0	3	16	12	14	2	0
0	0	4	16	16	2	0	0
0	3	16	8	10	13	2	0
0	1	15	1	3	16	8	0
0	0	11	16	15	11	1	0

one feature vector, x

0
0
9
14
⋮
1
0

64 numbers in all

the same picture as before, with 64 axes instead of 2

Writing it down

The same table, in the notation used for the rest of the course

A dataset D of n samples:

$$D = \{ (x_1, y_1), (x_2, y_2), \dots, (x_n, y_n) \}$$

What each symbol stands for

$$D = \{ (x_1, y_1), (x_2, y_2), \dots, (x_n, y_n) \}$$

x_i — the feature vector of sample i : everything we measured about it

y_i — the target of sample i : the answer for that one sample

n — how many samples there are

And this is the whole goal

Everything else is a detail of how

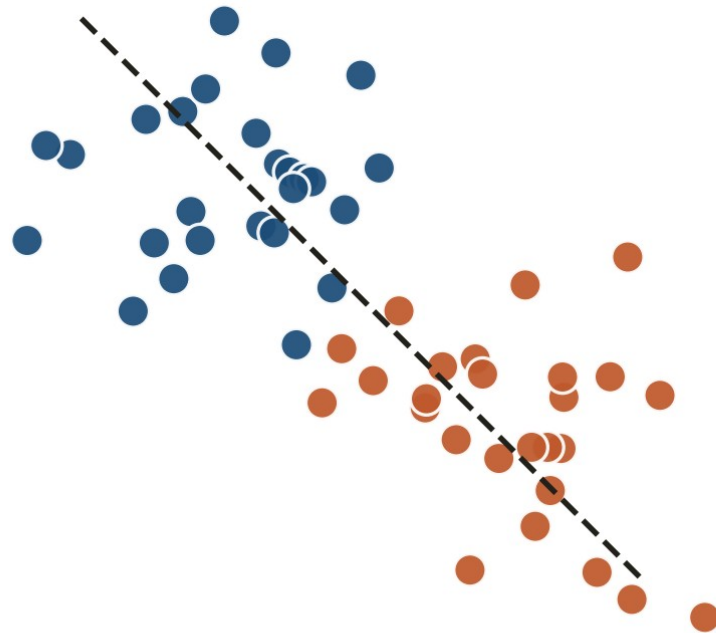
Find a function f such that $f(x) \approx y$

for samples we have never seen.

Every model in this course — every one of them — is a different answer to the question of which functions f we are willing to consider, and how we choose among them.

With answers, and without

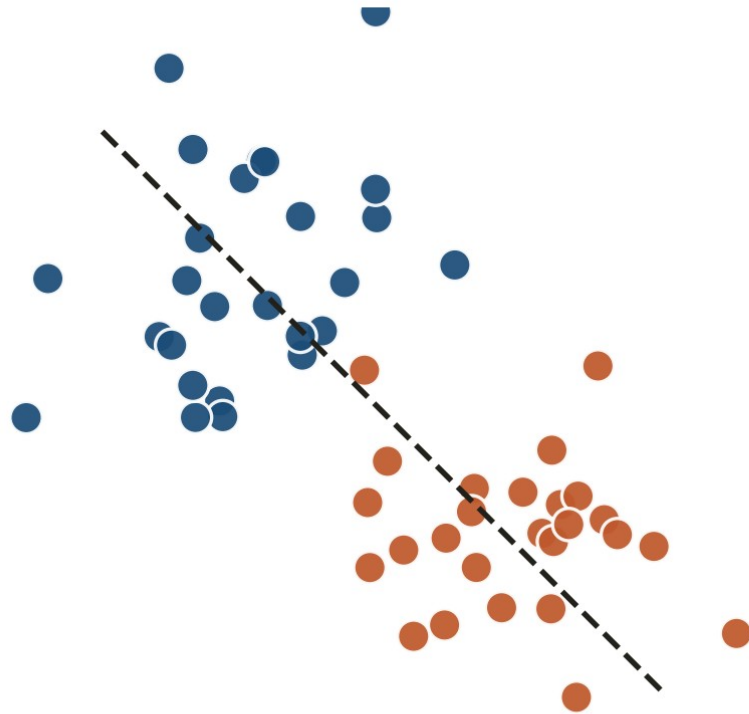
Supervised



every example carries its answer

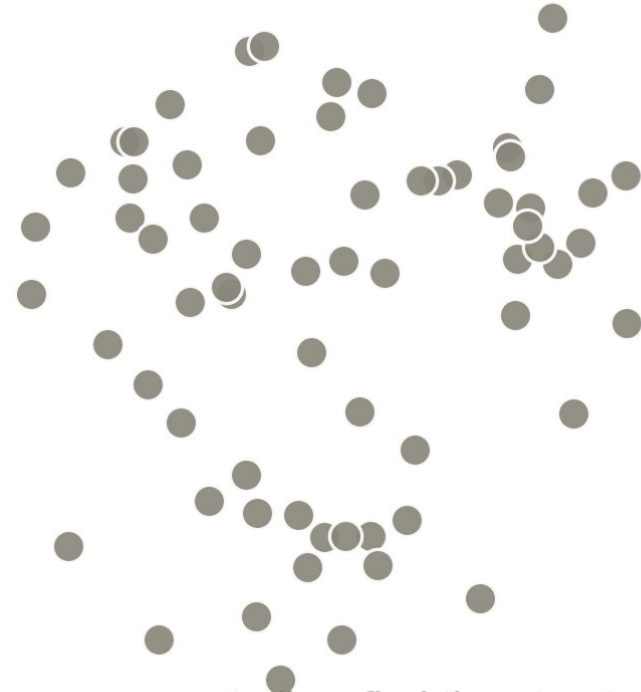
With answers, and without

Supervised



every example carries its answer

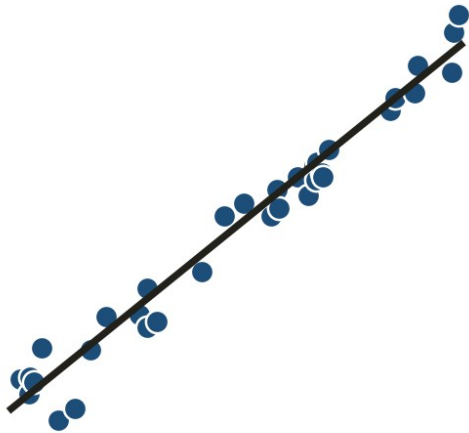
Unsupervised



no answers at all — find the structure

Four problems, and this course covers all of them

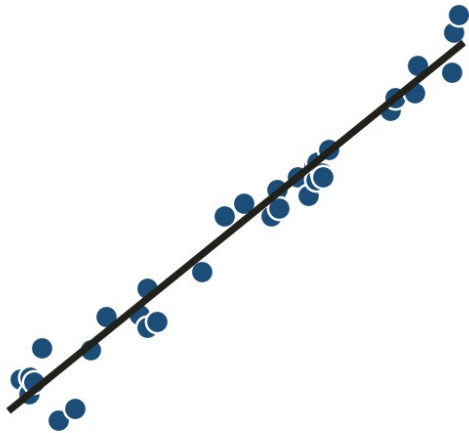
Regression



predict a number

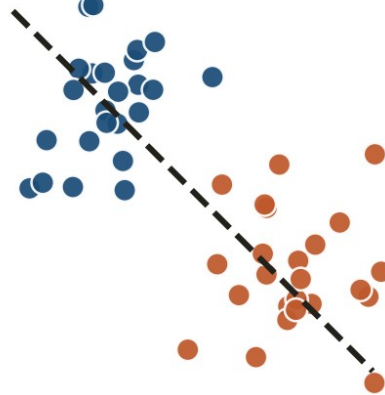
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Regression



predict a number

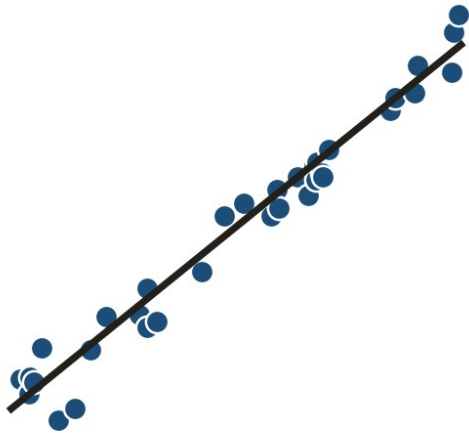
Classification



predict a category

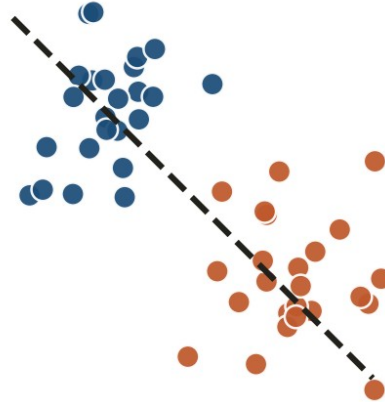
Four problems, and this course covers all of them

Regression



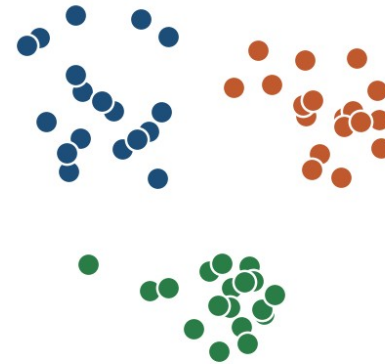
predict a number

Classification



predict a category

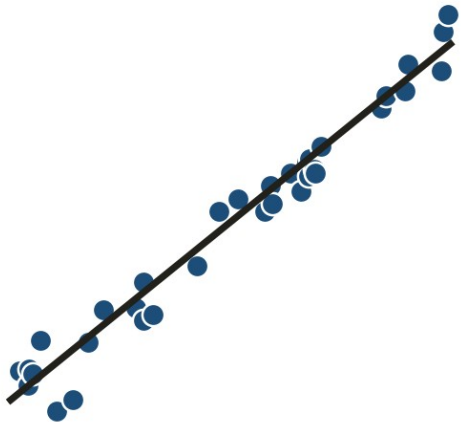
Clustering



group what belongs together

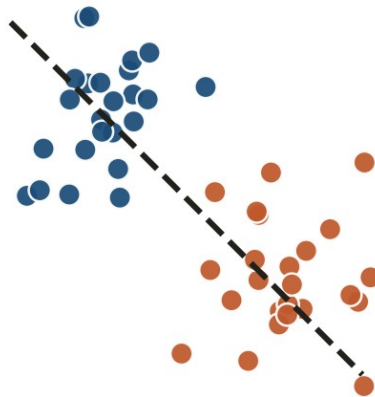
Four problems, and this course covers all of them

Regression



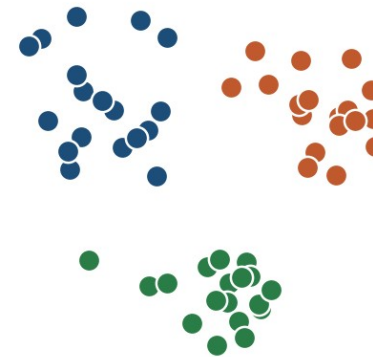
predict a number

Classification



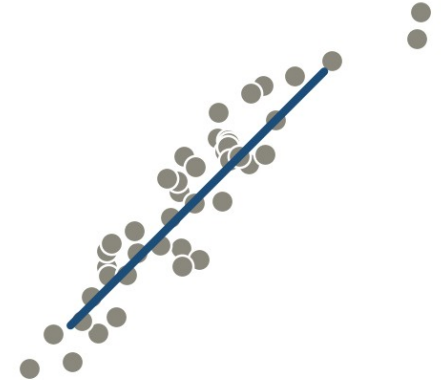
predict a category

Clustering



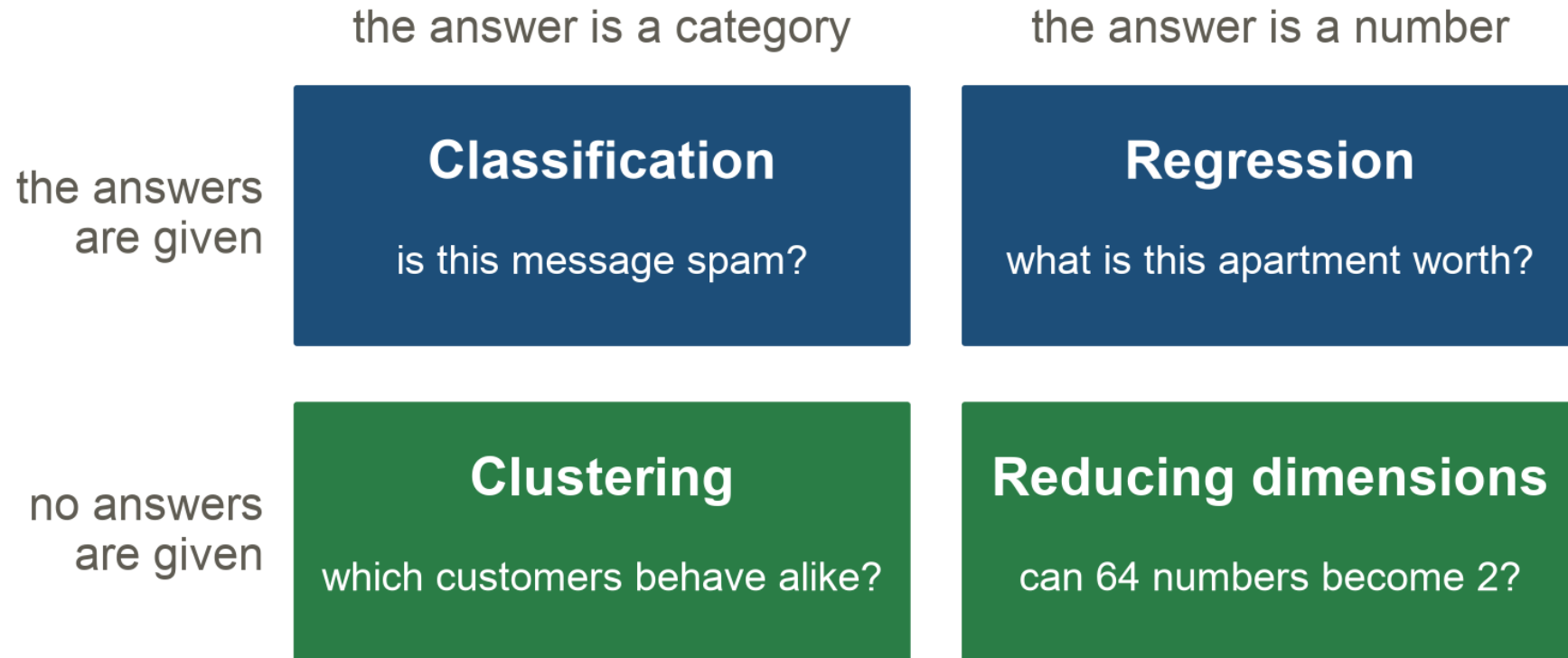
group what belongs together

Reducing dimensions



describe the data with fewer numbers

The same four, arranged by two questions



ask which box a new problem belongs in before naming the method

Learning is finding a function from features to target

- What goes in: samples, each described by features, written x .
- What comes out: a prediction, which we write $f(x)$.
- Supervised means we know y and can check the fit. Unsupervised means we cannot.
- From here on, when I say the model, I mean the function f .



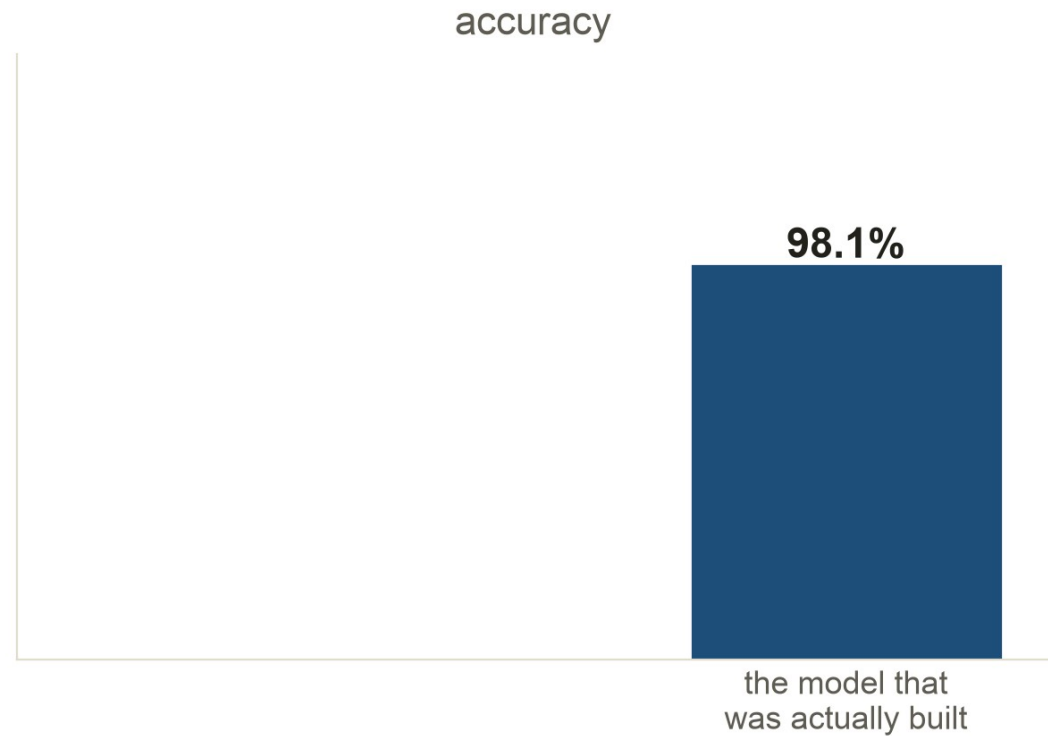
What does success mean?

Part 3 · measuring whether it worked

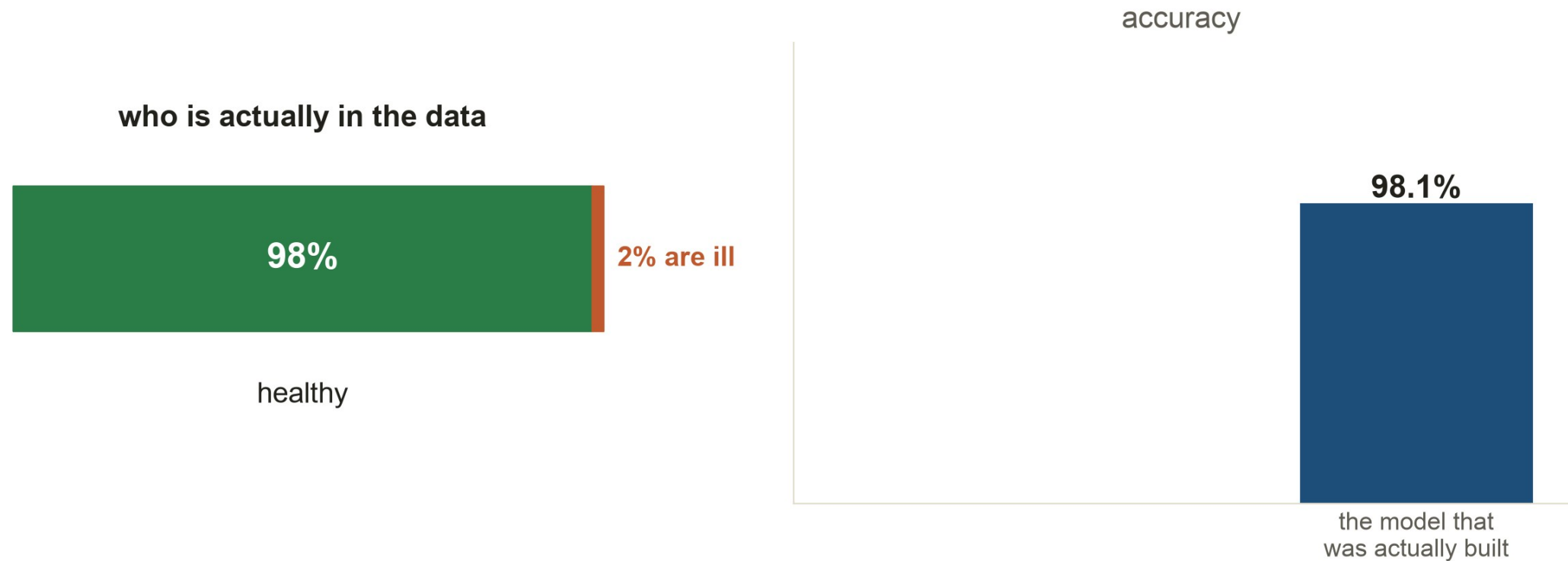
Is 98% accuracy good?

THE TRAP

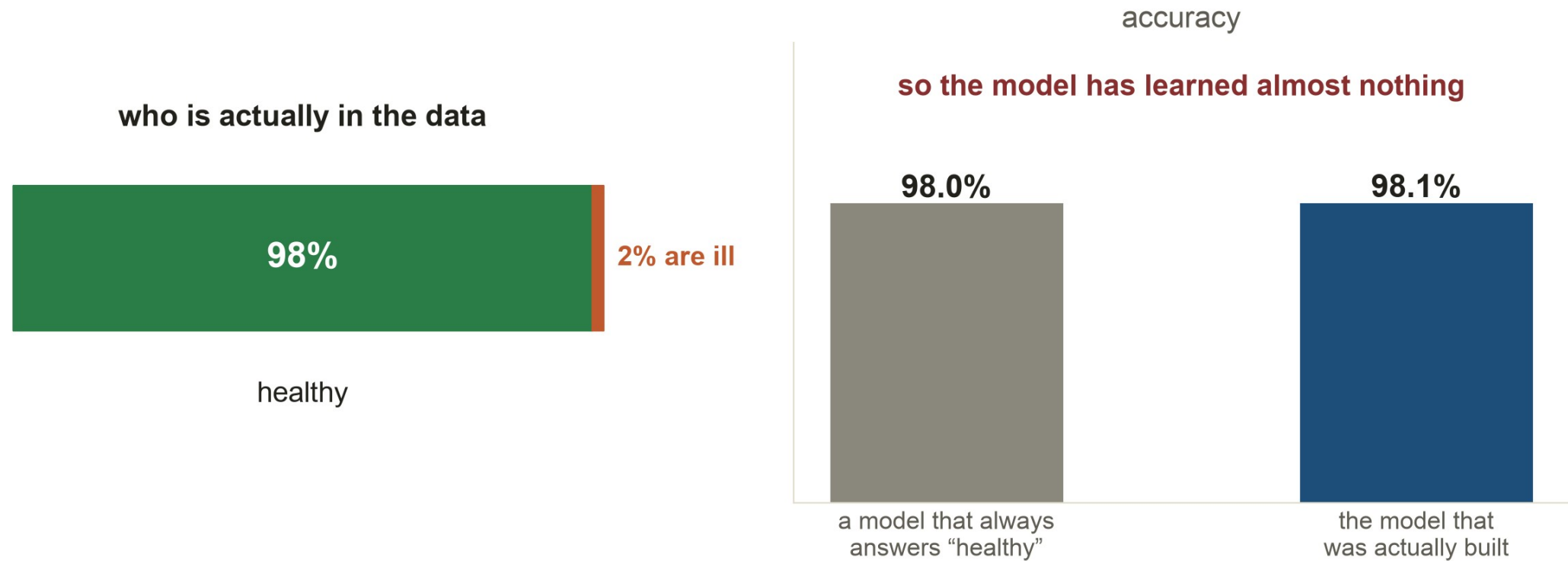
It depends entirely on the data



It depends entirely on the data



It depends entirely on the data



A number means nothing without the data it was computed on

Ninety-eight percent of what, measured on whom?

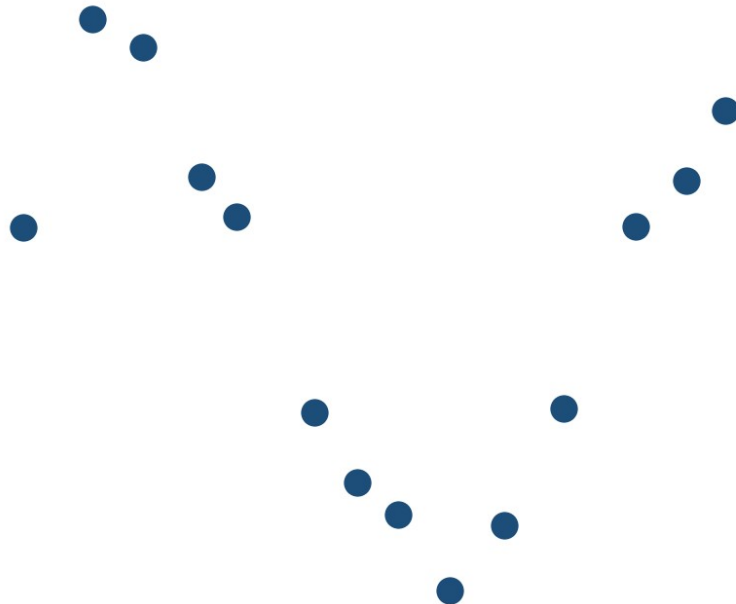
- We cannot judge a model in the abstract. We run it on data and measure.
- So every claim about performance is really a claim about a measurement.
- Which means the choice of data to measure on is not an administrative detail. It is the experiment.
- Which data, then?

Why can we not simply measure on the data we trained on?

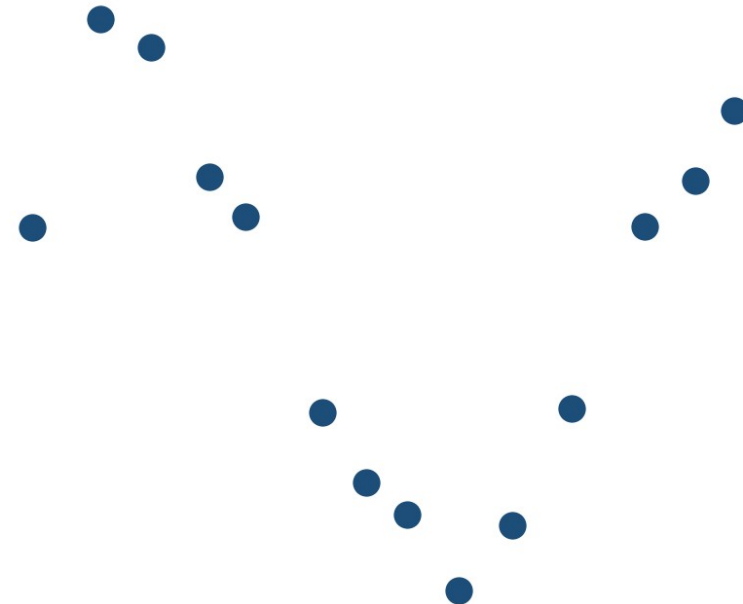
It has already seen the answers.

Two models, the same data

The same data

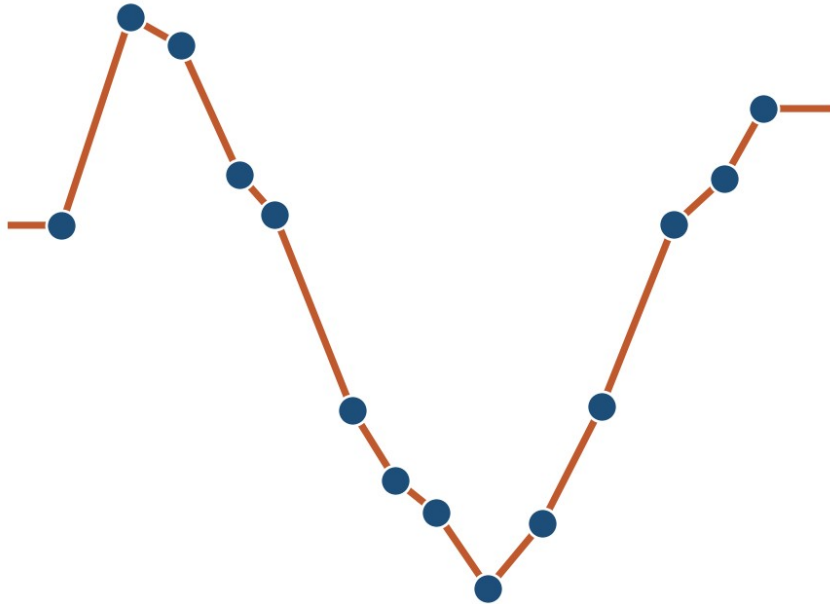


The same data

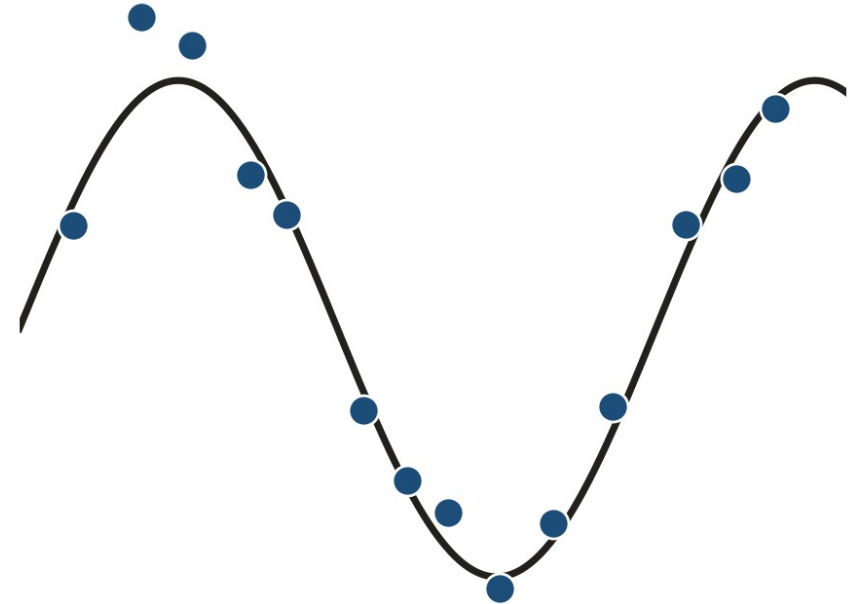


Two models, the same data

Memorizing

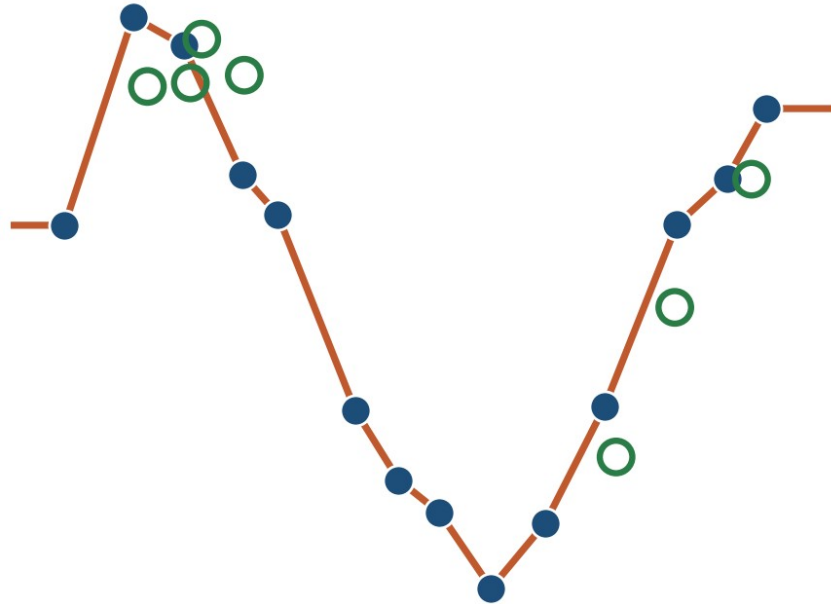


Generalizing



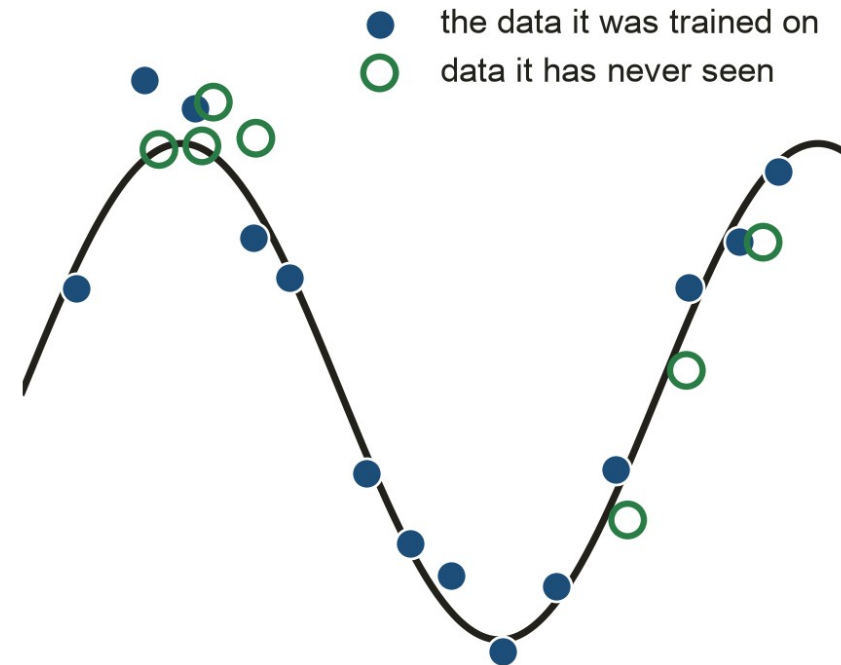
Two models, the same data

Memorizing



no error at all on the training data,
and badly wrong on the rest

Generalizing



a small error everywhere

Generalization, defined

This, and not accuracy on the training data, is the goal

A model generalizes when what it achieves on new data is close to what it achieved on the data it was trained on.

Memorizing: perfect on the training data, poor on everything else.

Generalizing: a small error everywhere, including on data it has never seen.

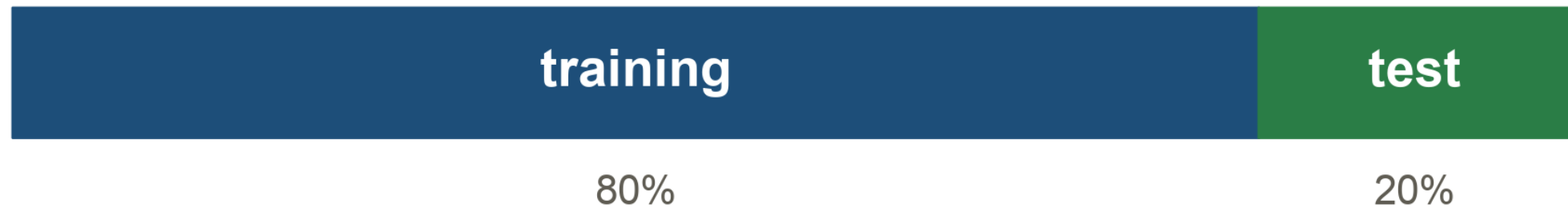
So we hold data back, on purpose



all of the labeled data

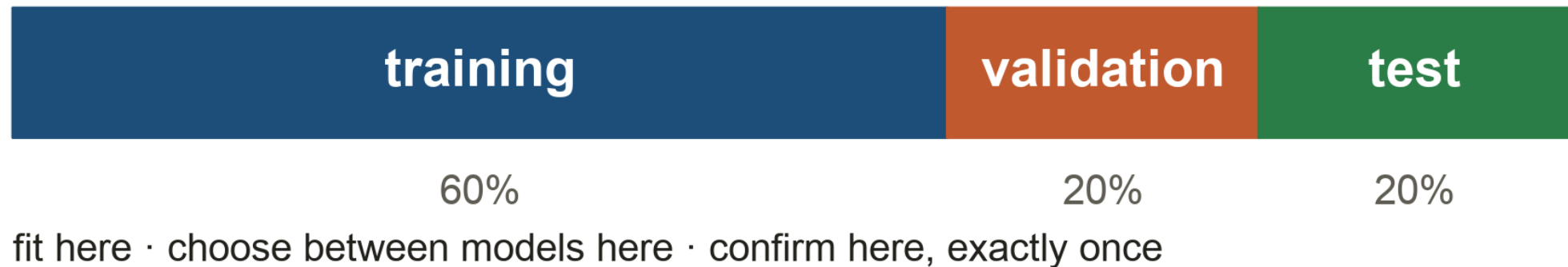
100%

So we hold data back, on purpose



fit the model on one part, judge it on the other

So we hold data back, on purpose



The rule about the test set

Why people call the test set sacred

Fit on the training data.

Choose between models on the validation data.

Look at the test data once, at the very end, and then stop.

Every time you look at the test data and change something afterwards, it becomes a little more like training data, and the number it produces becomes a little less honest.

TAKEAWAY

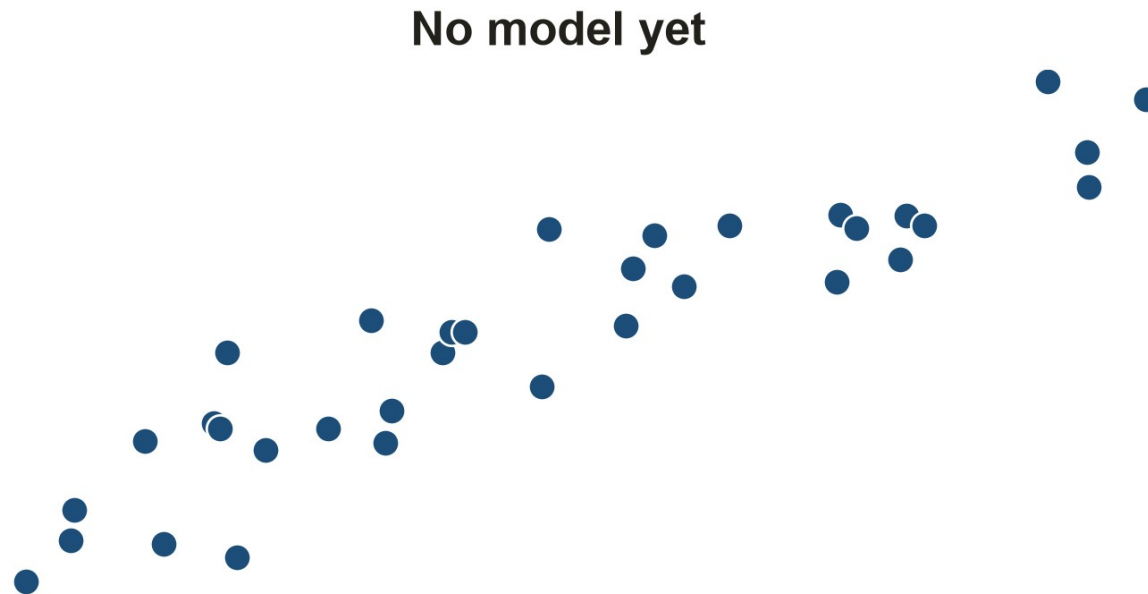
Success is performance on data the model has never seen

- We train on the past. What we care about is the future, and the future is unseen by definition.
- A model that only repeats what it was shown has not learned anything.
- This is why we hold data back before we start, rather than afterwards.

How complicated should a model be?

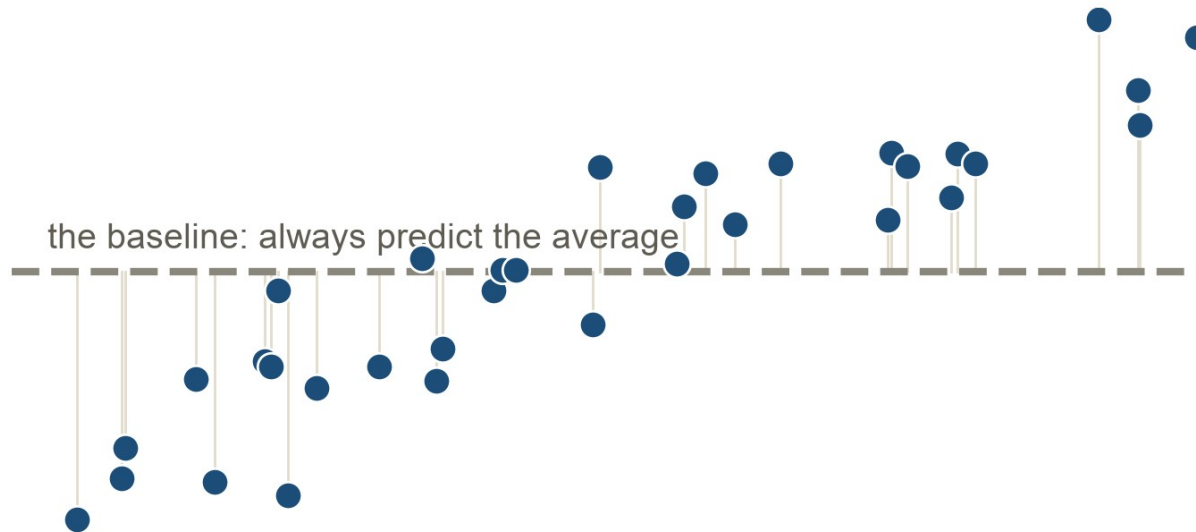
Part 4 · baselines, underfitting and overfitting

First, beat the stupidest possible rule

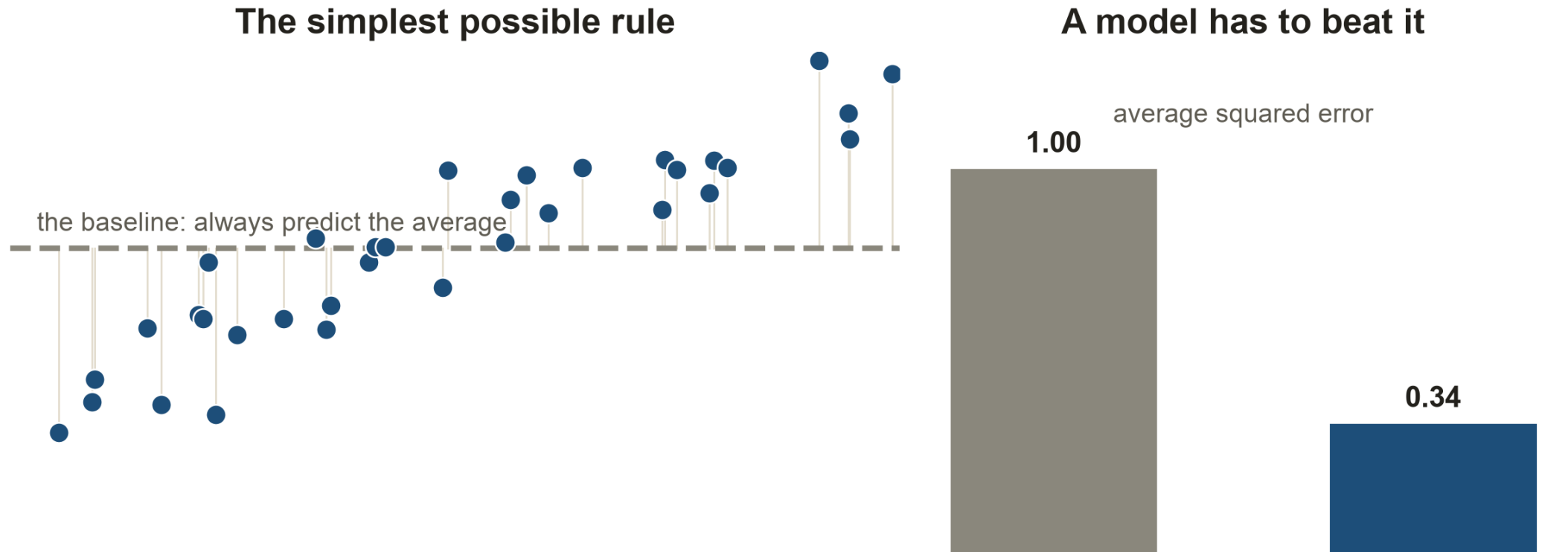


First, beat the stupidest possible rule

The simplest possible rule



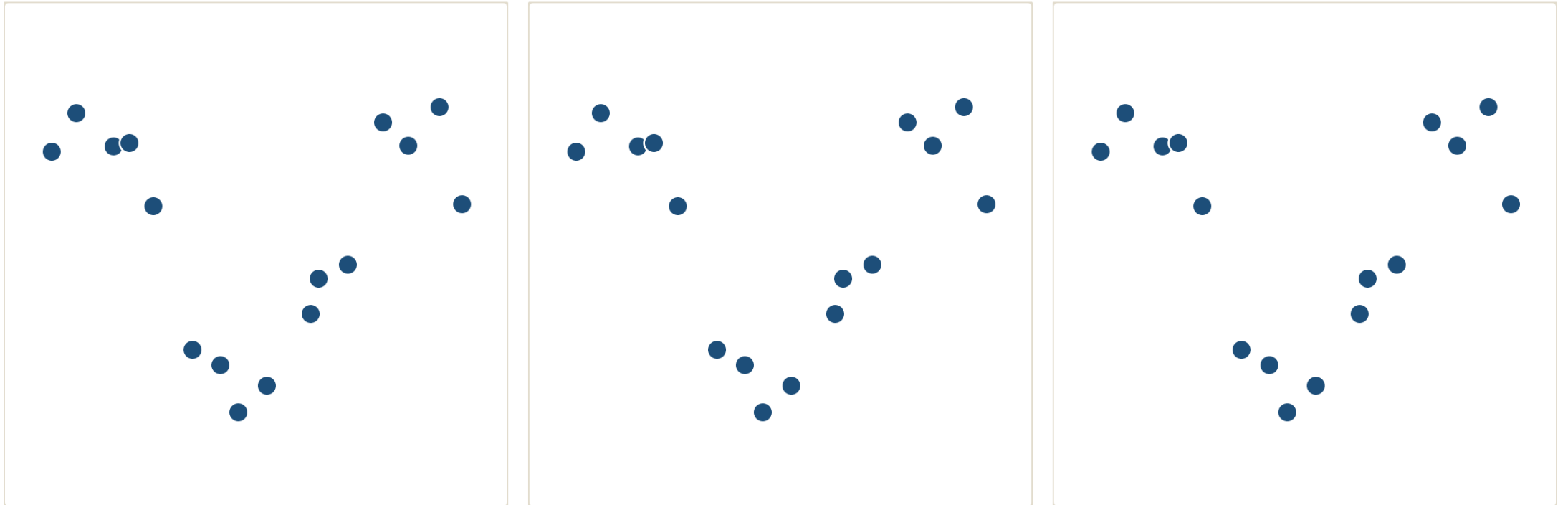
First, beat the stupidest possible rule



How complicated should the model be?

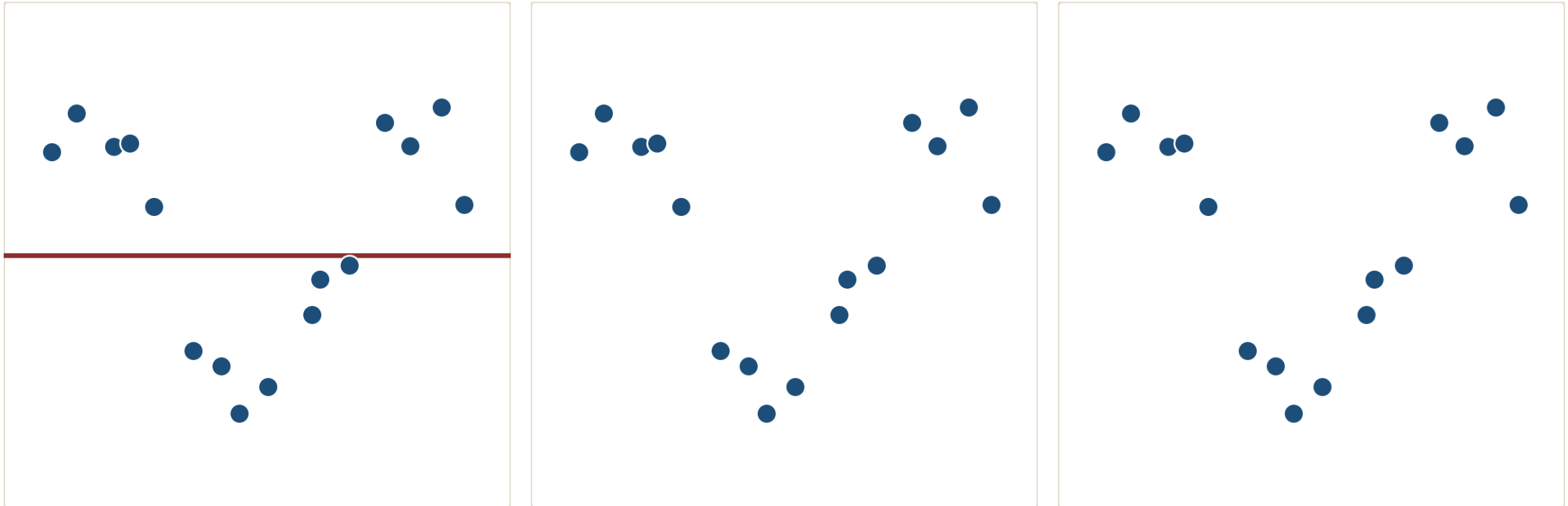
Too simple and it cannot follow the data. Too flexible and it follows the noise. Where is the line?

The two failures, and the answer between them



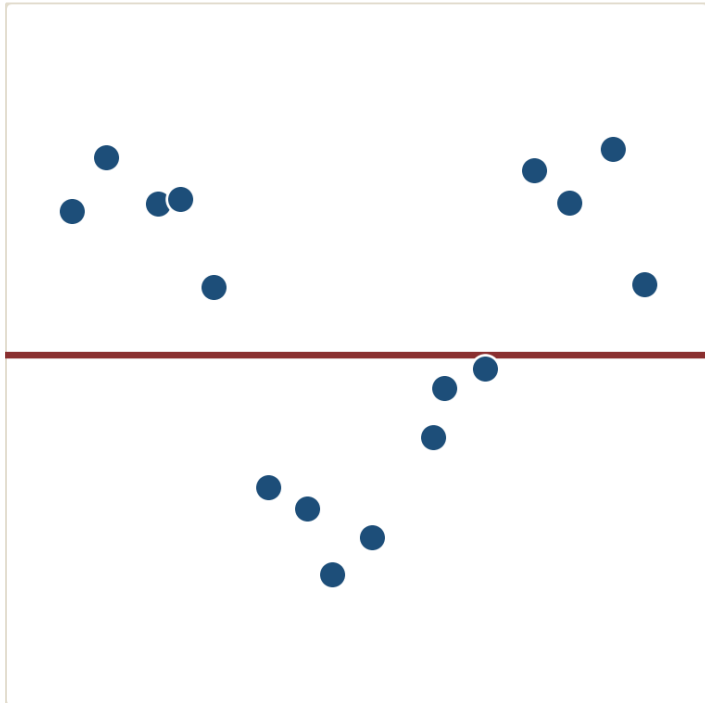
The two failures, and the answer between them

too simple

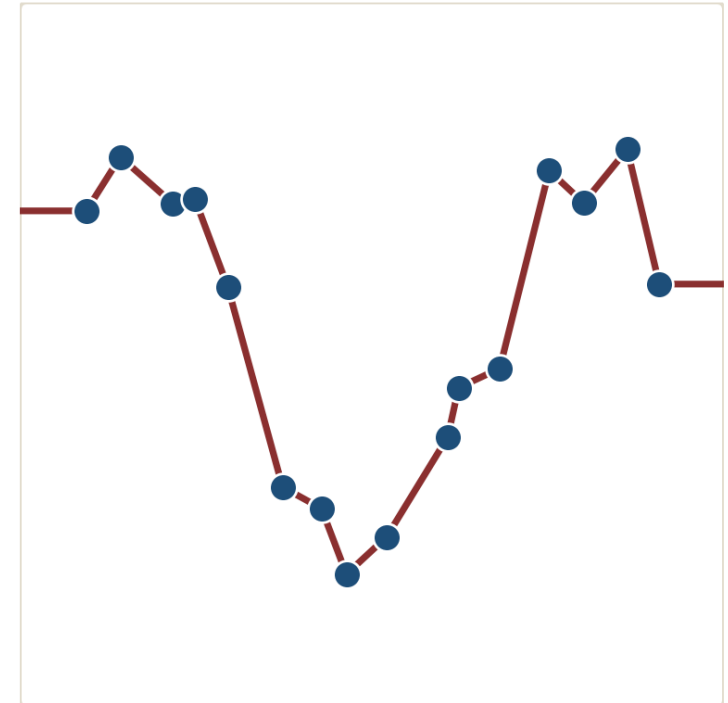


The two failures, and the answer between them

too simple

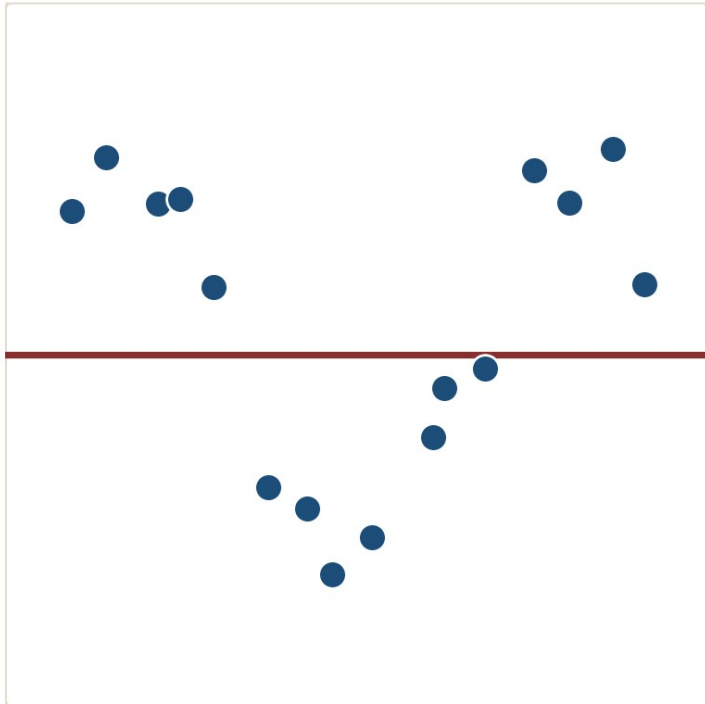


too flexible

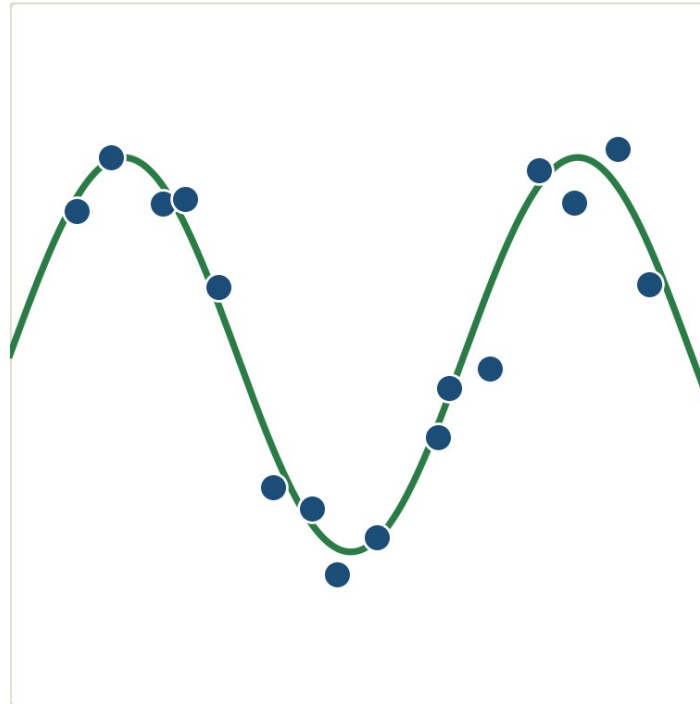


The two failures, and the answer between them

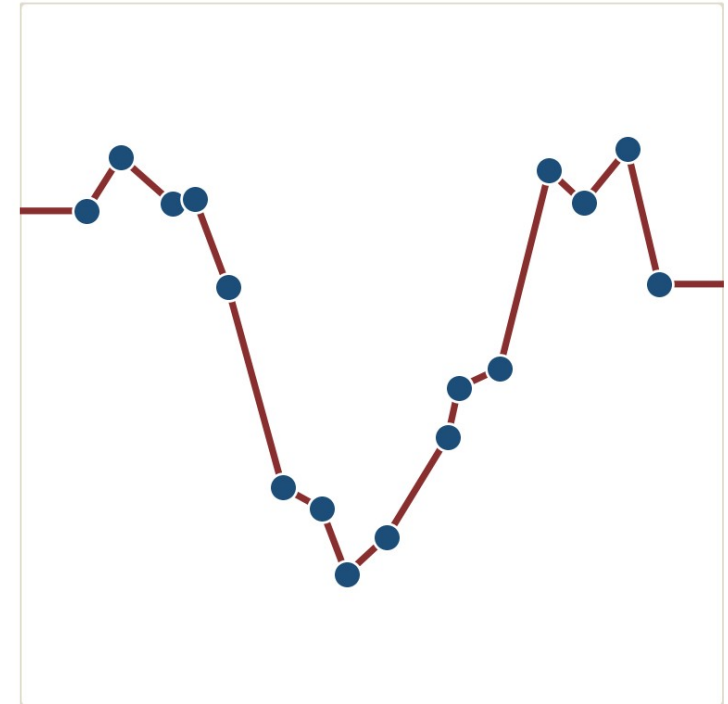
too simple



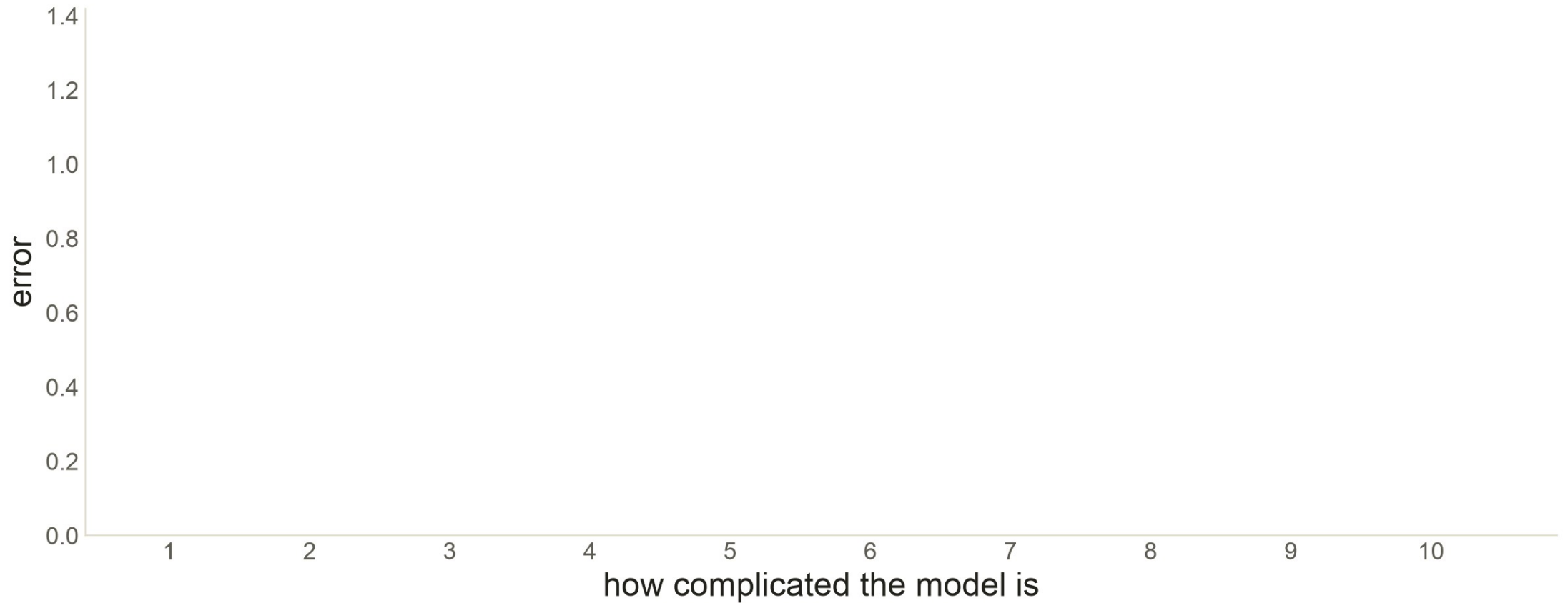
about right



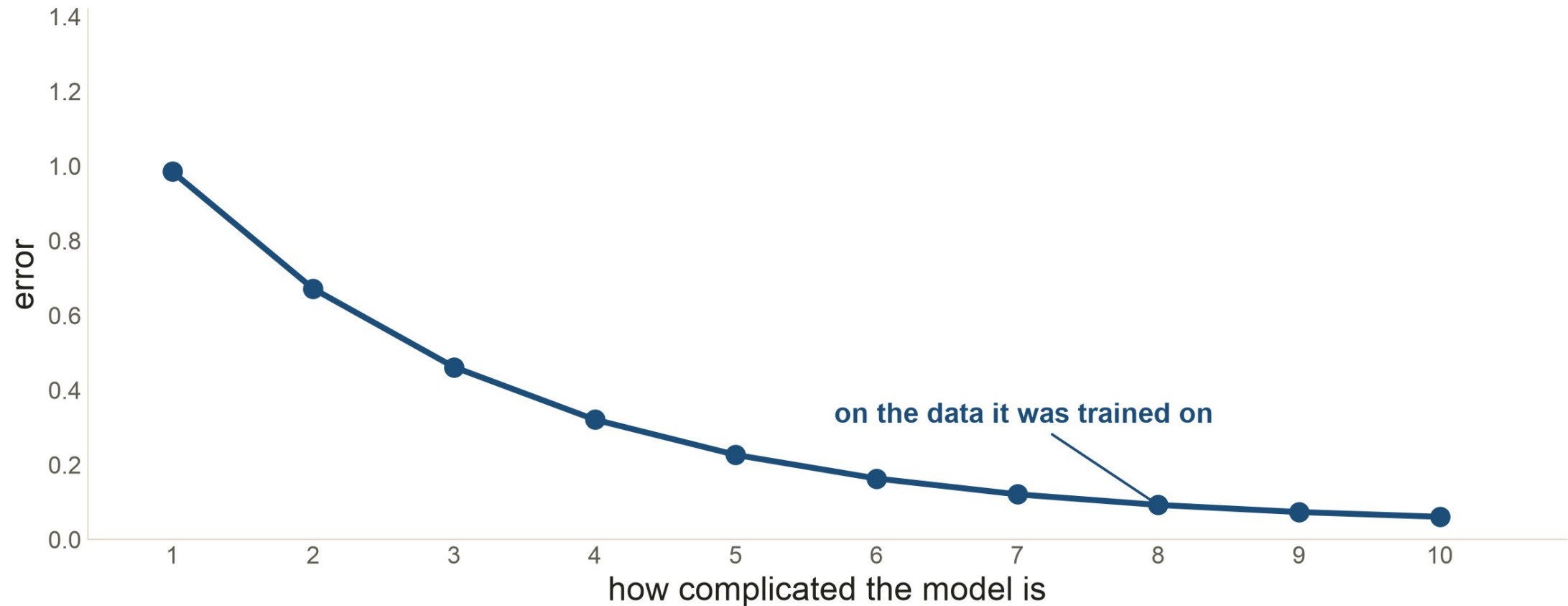
too flexible



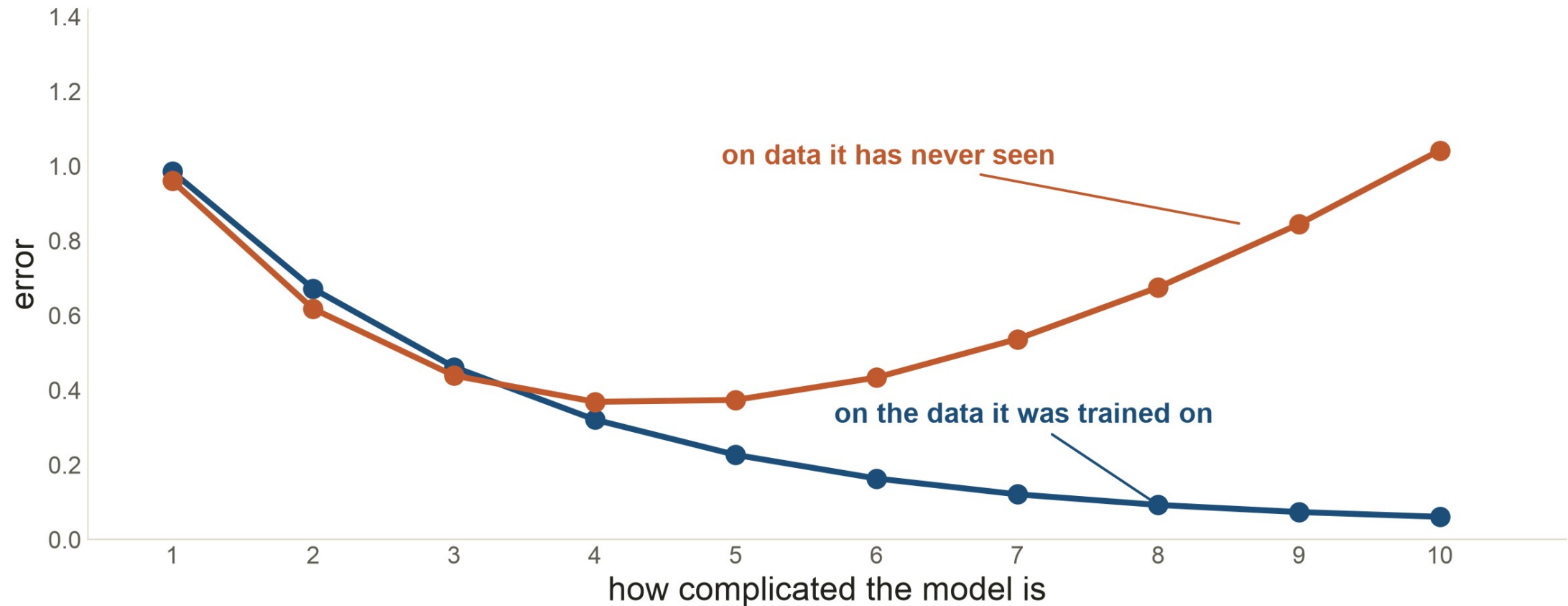
The curve that tells you where you are



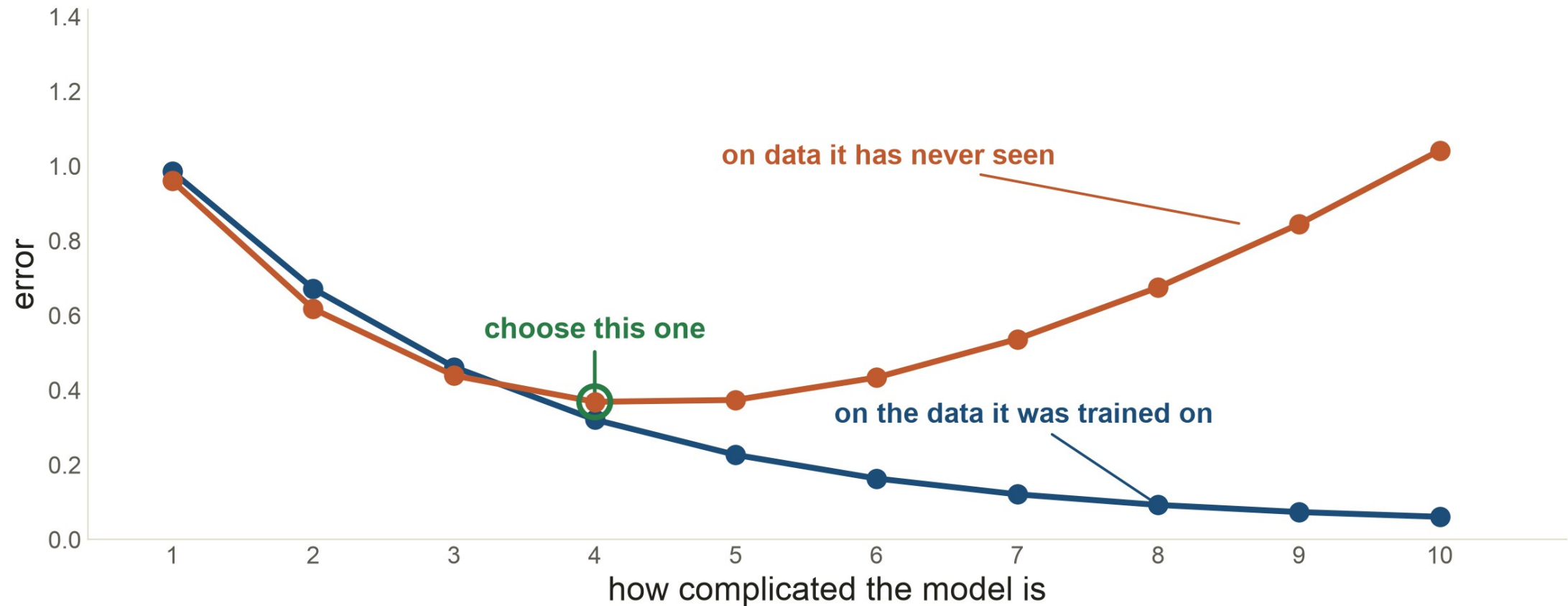
The curve that tells you where you are



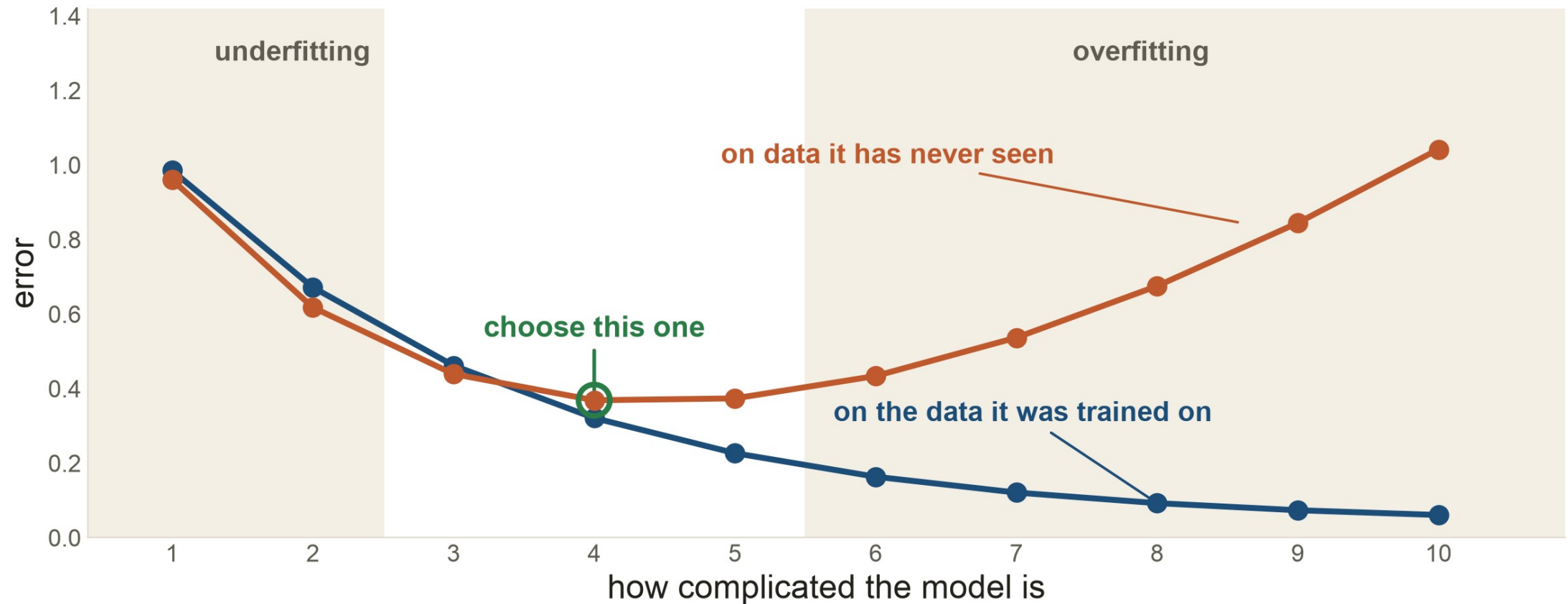
The curve that tells you where you are



The curve that tells you where you are



The curve that tells you where you are



Which model would you choose?

Two words you will use all semester

The same two failures, in the language everyone uses

Bias

The model is too rigid to represent the pattern. It is wrong in the same way every time, and more data will not help.

Variance

The model is flexible enough to chase the noise in this particular sample. Refit it on another sample of the same size and you get a noticeably different model.

Underfitting is too much bias. Overfitting is too much variance.

Choosing a model is an experiment, not a preference

- Fit on the training data, decide on the validation data, confirm once on the test data.
- Too much bias and too much variance are the two ways to fail, and they need opposite remedies.
- The validation curve is how you find out which one you are suffering from.



A good score can still be wrong.

Part 5 · data leakage

What data leakage is

Information from somewhere it should not have come from

Leakage is information reaching the model that would not be available when the prediction actually has to be made.

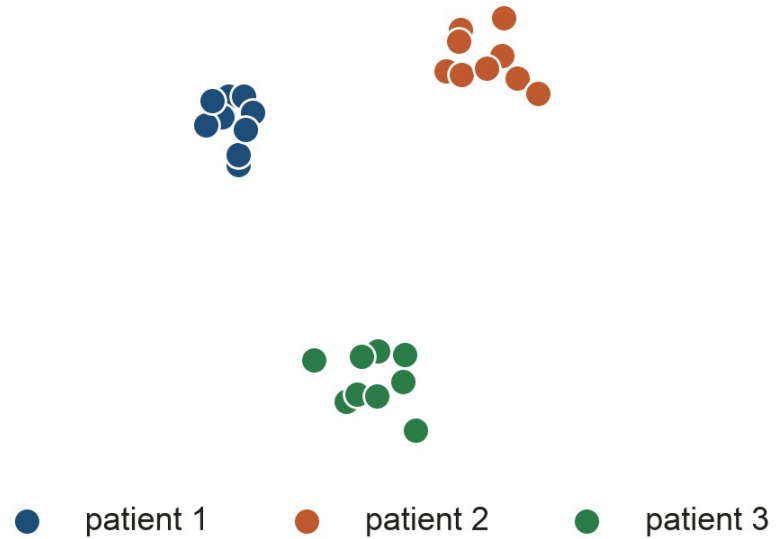
It always inflates the score, never deflates it.

It is invisible in the number. The number simply looks good.

CASE ONE

The same patient on both sides of the split

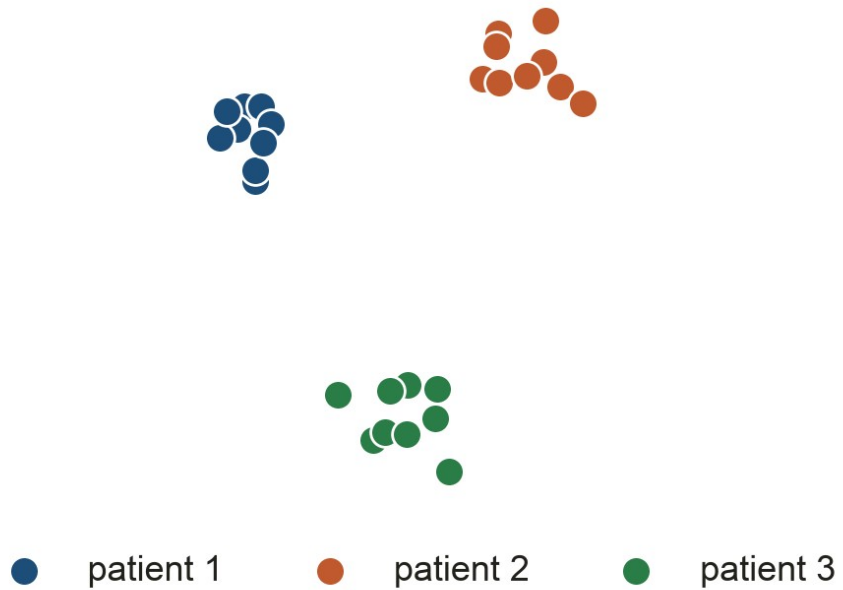
What the data really are



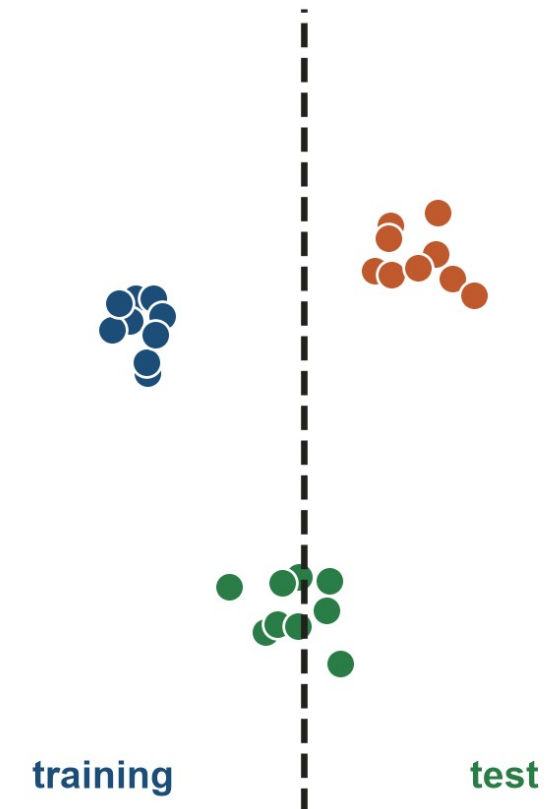
CASE ONE

The same patient on both sides of the split

What the data really are



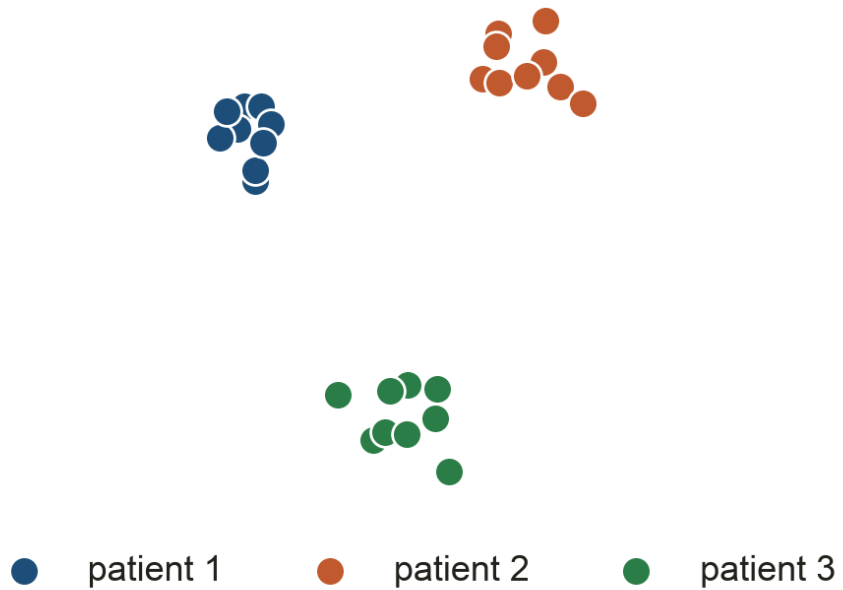
A split made at random



CASE ONE

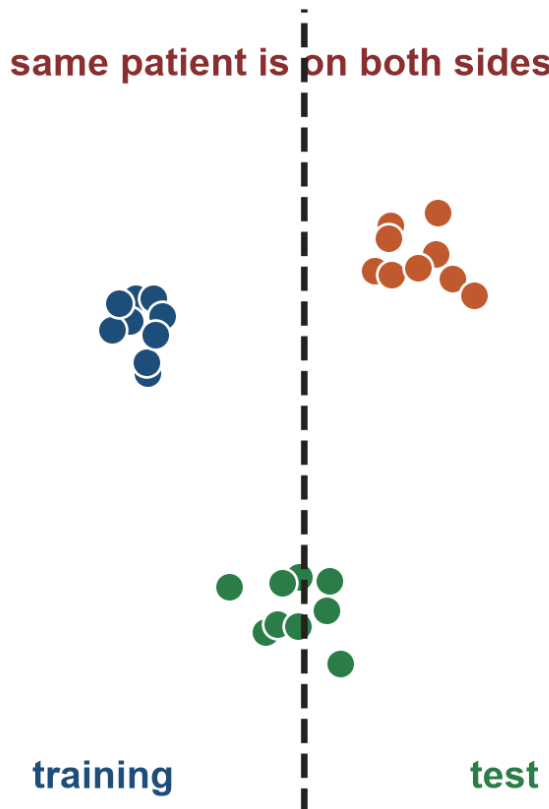
The same patient on both sides of the split

What the data really are



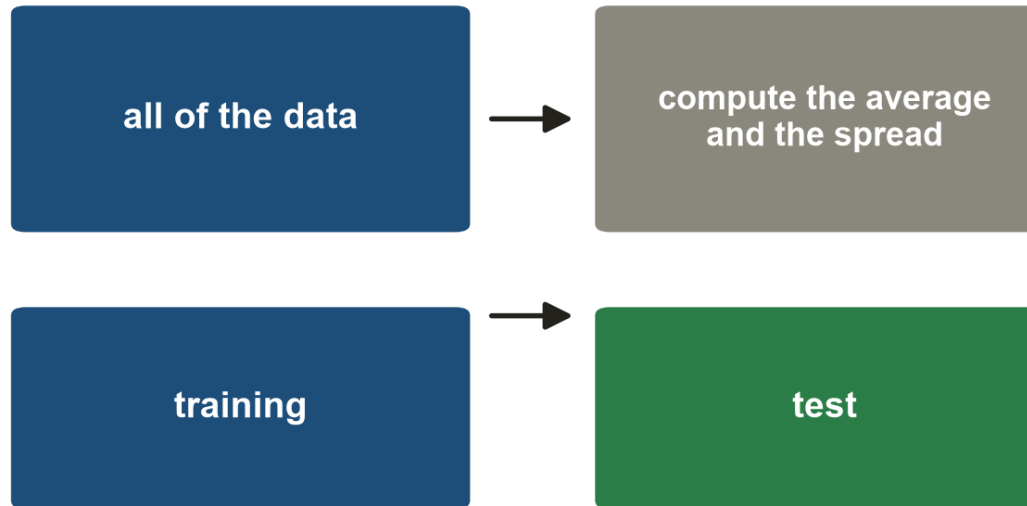
A split made at random

the same patient is on both sides



Preparing the data before splitting it

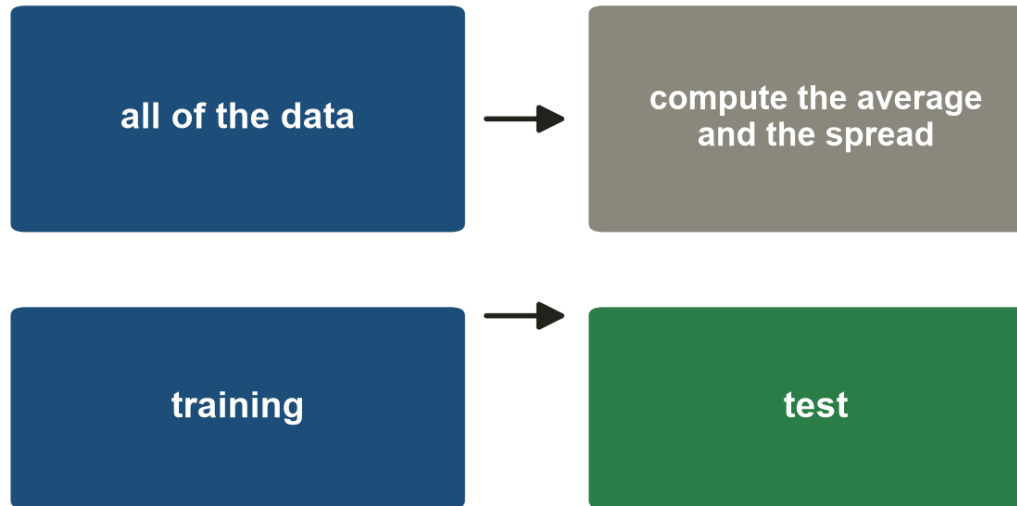
Wrong: scale first, split afterwards



those numbers already describe the test data

Preparing the data before splitting it

Wrong: scale first, split afterwards



those numbers already describe the test data

Right: split first, scale afterwards



nothing about the test data reaches the model

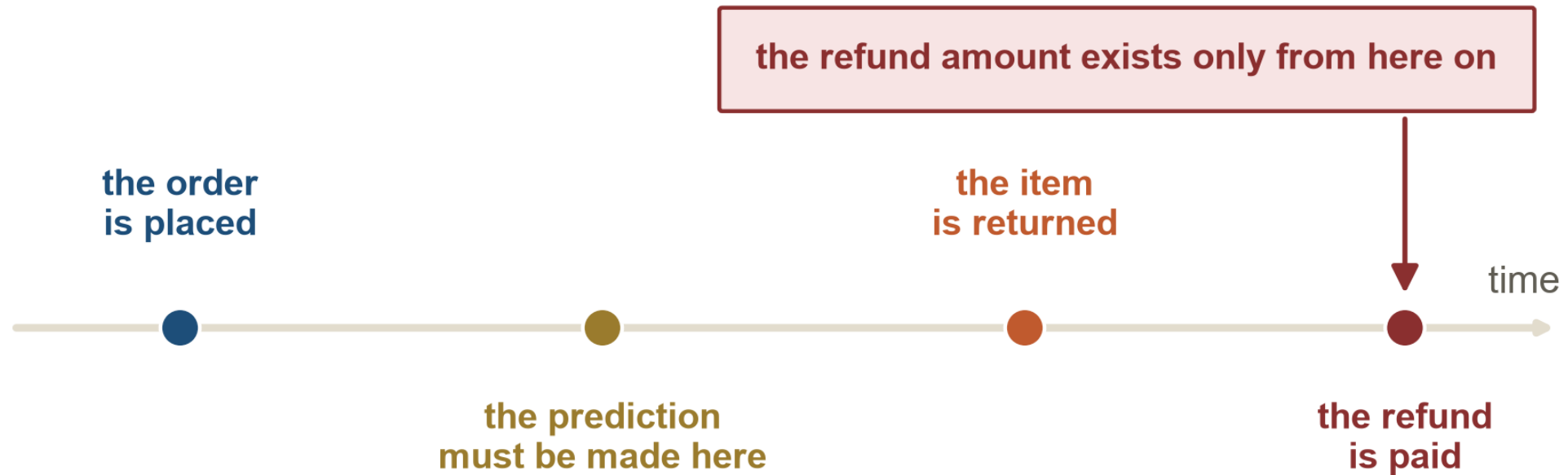
CASE THREE

A feature that did not exist yet



CASE THREE

A feature that did not exist yet



Using it as a feature means predicting the return from the refund that the return itself caused.

Would you trust this number?

Ask it about every result you see for the rest of the semester, including your own.

The design of the evaluation is part of the experiment

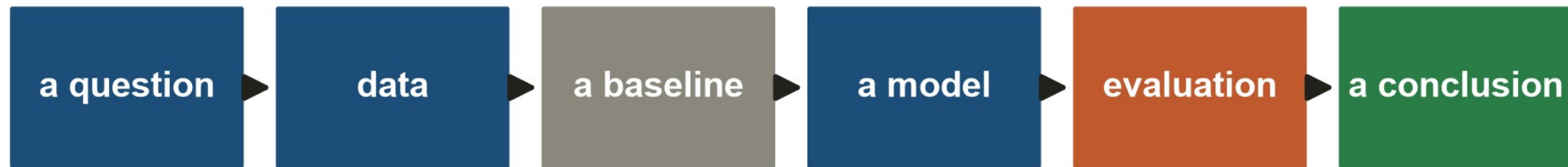
- A good score from a badly designed evaluation proves nothing at all.
- Two questions catch most of it: could this information have been available at prediction time, and did any step of the procedure see the test data?
- Suspicion of your own results is a skill. This course spends ten weeks training it.



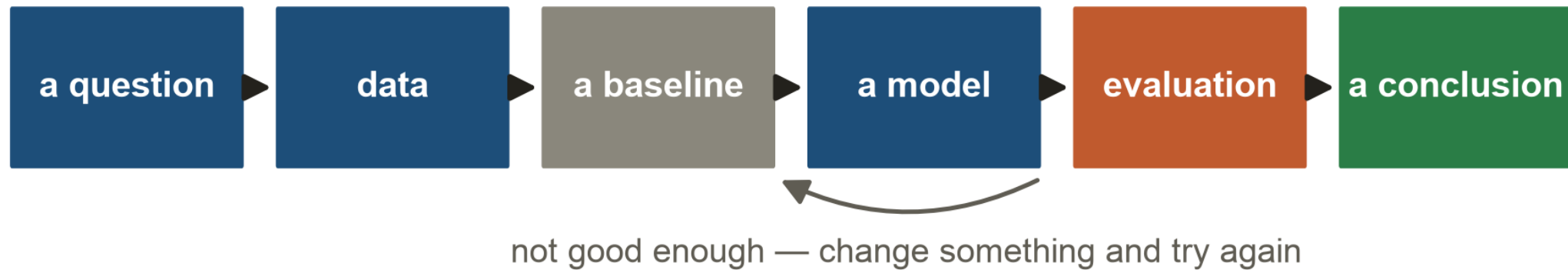
Putting it together

Part 6 · what an experiment actually looks like

Machine learning is a loop, not a function call



Machine learning is a loop, not a function call



Which is the loop you met at the start of the lecture



- **Experiment**
what you actually ran
- **Decision**
the choice you made, and why
- **Evidence**
the plot, table or number that supports it
- **Interpretation**
what that evidence means
- **Counterfactual**
what would change if one thing were different

notice what is not in it anywhere: fitting the model is not one of the five steps

What this course is not

It is not a tour of library functions. The implementation lives in the practical sessions. Here we work on when to use something, why it works, and whether to believe the result.

Three questions you can now answer

Answer them in your own words, not in mine

- 1. What does it mean for a machine to learn?**
- 2. What does the machine receive, and what does it produce?**
- 3. Why is training performance not the goal, and what is?**

This week

- Practical Session 1 — designing an experiment, and the simplest forms of leakage. It is not graded, and it is the training ground for both examinations.
- Read the syllabus and the policies page on the course website.
- Homework begins in Week 2. The next lecture is Learning a Function, on regression.

Machine Learning

Lecture 1 — What Does It Mean to Learn?

Questions?

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